Psychiatric Predictors of Violent Traumatic Brain Injury

Results from the TBIMS Study

In this study, we estimated the association between psychiatric illnesses/indices and TBI aetiology using pubically available data from the the Traumatic Brain Injury Model Systems National Database.

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Libraries and data

Packages

First load all necessary packages.

```
#install.packages("remotes")
#remotes::install_git("https://codeberg.org/hrbrmstr/hrbrthemes.git")
```

```
if (!require("pacman")) install.packages("pacman")

library(pacman)
pacman::p_load(
  remotes, # used to intstall R packages from remote/local repositories, eg, GitHub
  styler, # formats code automatically
  readxl, # read Excel files
  DT, # interface for DataTables
  kableExtra, # scrollable tables
  tidyverse, # data wrangling
  labelled, # handling labelled variables
  devtools, # package development
  modelsummary, # create tables and plots for descriptive statistics
  estimatr, # calculates difference in means with uncertainty estimates (modelsummary
  ggplot2, # data visualisations and plots
  ggridges, # ggplot2 extension for ridgeline plots
  cowplot, # ggplot add on
  MetBrewer,                                                                       ①
  hrbrthemes, # theme components for ggplot2
  tinytable, # editing/formatting tables
  knitr, # report generation
  gt, # presentation-level tables
  tibble, # utilities for handling tibbles
  naniar, # missing data
  VIM, # visualise missing data
  faraway, #ilogit() function
  sjPlot, # plotting ORs
  gtsummary, # presentation of regression analyses
  rms, # assessing the linearity assumption with restricted cubic splines
  reshape, # for the melt() function
  purrr, # for functions and vectors
  haven, # missing data MCAR
  finalfit, # missing data MCAR
  mice, # multiple imputation
  miceadds # multiple imputation
)

# download custon functions from the "Introduction to Regression Methods for Public H
source("Functions_rmph.R")
```

① This is one of my favourite packages, MetBrewer. It allows you to utilise palettes inspired by works at the Metropolitan Museum of Art in New York. This Quarto file uses the Klimt palette.

Dataviz

Create a datavisualisation theme.

```
scale_color_brand <- function() {
  custom_palette <- MetBrewer::met.brewer(name = "Klimt", n=20, type="continuous")   ②
  scale_color_manual(values = custom_palette)
}

scale_fill_brand <- function(labels = NULL) {
  custom_palette <- MetBrewer::met.brewer(name = "Klimt", n=20, type="continuous")
  if (is.null(labels)) {
    scale_fill_manual(values = custom_palette)
  } else {
    scale_fill_manual(values = custom_palette, labels = labels)
  }
}

palette <- MetBrewer::met.brewer(name = "Klimt", n=20, type="continuous")

theme_brand <- function() {
  theme_ipsum() +                                                                    ③
    theme(
      plot.title = element_text(color = "black", size = 14, face = "bold"),
      plot.subtitle = element_text(margin = margin(b = 20)),
      axis.text.x = element_text(size = 7),
      axis.text.y = element_text(size = 7),
      axis.title.x = element_text(hjust = 0.5),  # center x-axis label
      axis.title.y = element_text(hjust = 0.5),  # center y-axis label
      plot.background = element_rect(fill = "gray97", color = NA),
      plot.caption = element_text(hjust = 0)
    )
}
```

② The n function allows you to extend the number of values so the colour palette can still be used on graphs with numerous categorical values.

③ You can change the font using the hrbrthemes package. I don't have the other fonts installed, and its a bit of a pain to install them, hence why I used theme_ipsum.

Import data

Then, import the the publicly available de-identified data from TBIMS, and save the .csv file to a R studio project (no need to setwd).

```
data <- read_csv("data/input/TBIMSForm1_Public_20250812.csv")
data_long <- read_csv("data/input/TBIMSForm2_Public_20250812.csv")
```

Data linkage and wrangling

Data linkage

Now lets link Form 1 and Form 2 using union/vertical joins. Data is in the long format.

```
# create baseline rows from baseline data
baseline_long <- data %>%
  transmute(
    Mod1Id,
    Mod2Id = NA,
    FollowUpPeriod = 0,
    AGENoPHIF = AGENoPHI,
    EduYearsF = EduYears,
    EDUCATIONF = EDUCATION,
    EMPLOYMENTF = EMPLOYMENT,
    EarnF = Earn,
    RURALF = RURALadm,
    CombinedDRSF = DRSa,
    FIMTOTF = FIMTOTA,
    FIMCOGF = FIMCOGA,
    DRINKCatF = DRINKCat,
    PROBLEMUseF = PROBLEMUse,
    DrugsF = Drugs,
    SuicideF = Suicide,
    TBIInjuryF = TBIInjury,
    TBIRepInjuryF = TBIRepInjury,
    DeathCause1F = DeathCause1,
    DeathCause2F = DeathCause2,
    DeathECodeF = DeathECode
  ) # `transmute()` creates NEW columns and DROPS all other columns (unlike `mutate()` w

# create follow-up rows from follow-up data
followup_long <- data_long %>%
  select(Mod1Id, Mod2Id, FollowUpPeriod, AGENoPHIF, EduYearsF, EDUCATIONF, EarnF, EMPLOY

# combine baseline and follow-up (union/vertical joins)
longitudinal_data <- bind_rows(baseline_long, followup_long) %>%
  arrange(Mod1Id, FollowUpPeriod)

# add time-invariant demographics from baseline data
longitudinal_complete <- longitudinal_data %>%
  left_join(
    data %>%
      select(Mod1Id, SexF, RaceIndF, RaceBlkF, RacePIF, RaceAsnF, RaceWhtF,
             EthnicityF, CountryBirthF, INJYEAR, DEATHYEAR,
             Cause, GCS, SCI, CC_Hypertension, SpEd, SuicidePrior, PsyHosp, MntlEver),
    by = "Mod1Id"
  )
```

Check the unique number of individuals in the linked data.

```
longitudinal_complete %>%
  distinct(Mod1Id) %>%
  nrow()
```

```
[1] 19996
```

Data cleaning

Clean up the data. Ensure categorical variables are coded as factors as this makes it much easier to use in statistical models and graphs. Age, year, and time-to-event variables are coded as numeric. "Unknown" or missing data is coded as NA. Invalid values are also coded as NA.

Demographics

```
# age (AGENoPHIF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(AGENoPHIF_C = ifelse(AGENoPHIF < 16, NA, AGENoPHIF)) %>% # those with an incorr
  mutate(AGENoPHIF_C = ifelse(AGENoPHIF_C == 999, NA, AGENoPHIF_C)) %>% # a cleaned vers
  mutate(AGENoPHIF_C = ifelse(AGENoPHIF_C == 777, 89, AGENoPHIF_C)) # windorise those ag

# fill in missing age using baseline age + follow-up time
longitudinal_complete <- longitudinal_complete %>%
  group_by(Mod1Id) %>%
  mutate(
    baseline_age = first(AGENoPHIF_C[FollowUpPeriod == 0], default = NA_real_),
    AGENoPHIF_C = if_else(is.na(AGENoPHIF_C),
                          baseline_age + FollowUpPeriod,
                          AGENoPHIF_C)
  ) %>%
  select(-baseline_age) %>%
  ungroup() %>%
  mutate(AGENoPHIF_C = as.numeric(AGENoPHIF_C))

# sex (SexF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(SexF = as.factor(SexF)) %>%
  mutate(SexF_C = ifelse(SexF == 99 | is.na(SexF), NA, SexF)) %>%
  mutate(SexF_C = as.factor(SexF_C))

# race (RaceCat)
# the original `Race` variable (added in 1989) was archived in 2023 and is therefore not
## categorise race
longitudinal_complete <- longitudinal_complete %>%
  mutate(
    RaceIndF_C = ifelse(RaceIndF > 1, NA, RaceIndF),
    RaceBlkF_C = ifelse(RaceBlkF > 1, NA, RaceBlkF),
    RacePIF_C = ifelse(RacePIF > 1, NA, RacePIF),
    RaceAsnF_C = ifelse(RaceAsnF > 1, NA, RaceAsnF),
    RaceWhtF_C = ifelse(RaceWhtF > 1, NA, RaceWhtF)
  ) %>%
  mutate(
    RaceCount =  RaceIndF_C + RaceBlkF_C + RacePIF_C + RaceAsnF_C + RaceWhtF_C,
    RaceCat = case_when(
      RaceCount == 1 & RaceWhtF_C == 1 ~ 1, # white
      RaceCount == 1 & RaceBlkF_C == 1 ~ 2, # black
      RaceCount == 1 & RacePIF_C == 1 ~ 3, # PIF
      RaceCount == 1 & RaceAsnF_C == 1 ~ 4, # Asian
      RaceCount == 1 & RaceIndF_C == 1 ~ 5, # AIAN
      RaceCount > 1 ~ 6,  #mixed race category
      is.na(RaceBlkF_C) & is.na(RacePIF_C) & is.na(RaceAsnF_C) & is.na(RaceIndF_C) & !is
      !is.na(RaceBlkF_C) & is.na(RaceWhtF_C) & is.na(RacePIF_C) & is.na(RaceAsnF_C) & is
    !is.na(RacePIF_C) & is.na(RaceWhtF_C) & is.na(RaceBlkF_C) & is.na(RaceAsnF_C) & is.n
    !is.na(RaceAsnF_C) & is.na(RaceWhtF_C) & is.na(RaceBlkF_C) & is.na(RacePIF_C) & is.n
    !is.na(RaceIndF_C) & is.na(RaceWhtF_C) & is.na(RaceBlkF_C) & is.na(RacePIF_C) & is.n
    )
  ) %>%
  mutate(Race1 = case_when(
    RaceCat == 1 ~ 1, # white
    RaceCat == 2 ~ 2, # black
    RaceCat == 4 ~ 3, # asian
    RaceCat == 3 ~ 4, # NHPI
    RaceCat == 5 ~ 5, # AIAN
    RaceCat == 6 ~ 6 # mixed race
  )) %>%
  mutate(RaceCat = as.factor(RaceCat)) %>%
  mutate(Race1 = as.factor(Race1))

# ethnicity (EthnicityF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(EthnicityF = as.factor(EthnicityF)) %>%
  mutate(EthnicityF_C= ifelse(EthnicityF %in% c(66, 77, 99) | is.na(EthnicityF), NA, Eth
  mutate(EthnicityF_C = as.factor(EthnicityF_C))

# race_ethnicity
longitudinal_complete <- longitudinal_complete %>%
  mutate(
    race_ethnicity = case_when(
      # hispanic
      EthnicityF_C == 2 & Race1 == 1 ~ "Hispanic White",
      EthnicityF_C == 2 & Race1 == 2 ~ "Hispanic Black",
      EthnicityF_C == 2 & Race1 == 3 ~ "Hispanic Asian",
      EthnicityF_C == 2 & Race1 == 4 ~ "Hispanic NHPI",
      EthnicityF_C == 2 & Race1 == 5 ~ "Hispanic AIAN",
      EthnicityF_C == 2 & Race1 == 6 ~ "Hispanic Mixed Race",
      EthnicityF_C == 2 & is.na(Race1) ~ "Hispanic (Race Not Reported)",

      # non-hispanic
      EthnicityF_C == 1 & Race1 == 1 ~ "Non-Hispanic White",
      EthnicityF_C == 1 & Race1 == 2 ~ "Non-Hispanic Black",
      EthnicityF_C == 1 & Race1 == 3 ~ "Non-Hispanic Asian",
      EthnicityF_C == 1 & Race1 == 4 ~ "Non-Hispanic NHPI",
      EthnicityF_C == 1 & Race1 == 5 ~ "Non-Hispanic AIAN",
      EthnicityF_C == 1 & Race1 == 6 ~ "Non-Hispanic Mixed Race",
      EthnicityF_C == 1 & is.na(Race1) ~ "Non-Hispanic (Race Not Reported)",

      # race only
      is.na(EthnicityF_C) & Race1 == 1 ~ "White (Ethnicity Not Reported)",
      is.na(EthnicityF_C) & Race1 == 2 ~ "Black (Ethnicity Not Reported)",
      is.na(EthnicityF_C) & Race1 == 3 ~ "Asian (Ethnicity Not Reported)",
      is.na(EthnicityF_C) & Race1 == 4 ~ "NHPI (Ethnicity Not Reported)",
      is.na(EthnicityF_C) & Race1 == 5 ~ "AIAN (Ethnicity Not Reported)",
      is.na(EthnicityF_C) & Race1 == 6 ~ "Mixed Race (Ethnicity Not Reported)",

      # neither race or ethnicity reported
      TRUE ~ NA_character_
    )
  ) %>%

  # NIH reporting categories
  mutate(
    race_ethnicity_C = case_when(
      # Hispanic ALWAYS overrides race
      str_starts(race_ethnicity, "Hispanic") ~ "Hispanic or Latino",

      # to reduce missingness → Non-Hispanic other is a category
      str_detect(race_ethnicity, "Non-Hispanic \\(Race Not Reported\\)") ~ "Non-Hispanic

      # race-based NIH categories
      str_detect(race_ethnicity, "Asian") | str_detect(race_ethnicity, "NHPI") ~ "Asian
      str_detect(race_ethnicity, "Black") ~ "Black or AA",
      str_detect(race_ethnicity, "White") ~ "White",
      str_detect(race_ethnicity, "AIAN") ~ "AIAN",

      # If mixed and non-Hispanic → NIH would classify as "More than one race"
      str_detect(race_ethnicity, "Mixed") ~ "More than one race",

      TRUE ~ "Unknown"
    )
  ) %>%

  mutate(race_ethnicity_C = case_when(
    race_ethnicity_C == "White" ~ 1,
    race_ethnicity_C == "Black or AA" ~ 2,
    race_ethnicity_C == "Asian / NHPI" ~ 3,
    race_ethnicity_C == "Hispanic or Latino" ~ 4,
    race_ethnicity_C == "AIAN" ~ 5,
    race_ethnicity_C == "Non-Hispanic Other" ~ 6,
    race_ethnicity_C == "More than one race" ~ 7,
    race_ethnicity_C == "Unknown" ~ NA
  ),
  race_ethnicity_C = as.factor(race_ethnicity_C))

# rural (RURALF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(RURALF = as.factor(RURALF))

# country of birth (CountryBirthF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(CountryBirthF = as.factor(CountryBirthF)) %>%
  mutate(CountryBirthF_C= ifelse(CountryBirthF %in% c(77, 99) | is.na(CountryBirthF), NA
  mutate(CountryBirthF_C = as.factor(CountryBirthF_C))

# education (EduYearsF)
## categorise those with less than 10 years of school as those "still in school"
longitudinal_complete <- longitudinal_complete %>%
  mutate(EduYears10 = case_when(
    EduYearsF <= 10 ~ 1,
    EduYearsF == 11 ~ 2,
    EduYearsF == 12 ~ 3,
    EduYearsF == 13 ~ 4,
    EduYearsF == 14 ~ 5,
    EduYearsF == 15 ~ 6,
    EduYearsF == 16 ~ 7,
    EduYearsF == 17 ~ 8,
    EduYearsF == 18 ~ 9,
    EduYearsF == 19 ~ 10,
    EduYearsF == 20 ~ 11,
    EduYearsF == 21 ~ 12,
    EduYearsF == 666 ~ 666,
    EduYearsF == 999 ~ 999
  )) %>%
  mutate(EduYears10_C = if_else(EduYears10 %in% c(666, 999) | is.na(EduYears10), NA, Edu
  mutate(EDUCATION_C = case_when(
           EDUCATIONF == 1 | EDUCATIONF == 2 ~ 1, # in school
           EDUCATIONF == 2.5 | EDUCATIONF == 3 | EDUCATIONF == 3.5 ~ 2, # HS/GED
           EDUCATIONF == 5 ~ 3, # some college
           EDUCATIONF == 6 ~ 4, # associated degree
           EDUCATIONF == 7 ~ 5, # bachelors
           EDUCATIONF == 8 | EDUCATIONF == 9 ~ 6, # graduate degree
           EDUCATIONF == 4 | EDUCATIONF == 21 ~ NA, # other (due to the small number of
           EDUCATIONF == 999 ~ NA
           )) %>%
  mutate(EduYears10_C = as.factor(EduYears10_C)) %>%
  mutate(EDUCATIONF = as.factor(EDUCATIONF)) %>%
  mutate(EDUCATION_C = as.factor(EDUCATION_C))

# income (EarnF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(EarnF = as.factor(EarnF)) %>%
  mutate(EarnF_C = if_else(EarnF %in% c(666, 777, 999) | is.na(EarnF), NA, EarnF)) %>%
  mutate(EarnF_C = as.factor(EarnF_C))

# household income (HHIncomeF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(HHIncomeF = as.factor(HHIncomeF)) %>%
  mutate(HHIncomeF_C = if_else(HHIncomeF %in% c(66, 77, 99) | is.na(HHIncomeF), NA, HHIn
  mutate(HHIncomeF_C = as.factor(HHIncomeF_C))

# special education (SpEd)
## those with NA are assigned the value of "99 = Unknown"
longitudinal_complete <- longitudinal_complete %>%
  mutate(SpEd = as.factor(SpEd)) %>%
  mutate(SpEd_C= ifelse(SpEd %in% c(66, 77, 99) | is.na(SpEd), NA, SpEd)) %>%
  mutate(SpEd_C = as.factor(SpEd_C))

# employment (EMPLOYMENT)
## those with NA are assigned the value of "999 = Unknown"
longitudinal_complete <- longitudinal_complete %>%
  mutate(EMPLOYMENTF_C = ifelse(EMPLOYMENTF == 999 | is.na(EMPLOYMENTF), NA, EMPLOYMENTF
  mutate(EMPLOY1 = case_when(
    EMPLOYMENTF_C == 2 | EMPLOYMENTF_C == 3  ~ 2, # student (part or full time)
    EMPLOYMENTF_C == 5 ~ 1, # competitively employed
    EMPLOYMENTF_C == 9 ~ 3, # retired
    EMPLOYMENTF_C == 4 | EMPLOYMENTF_C == 7 | EMPLOYMENTF_C == 11 | EMPLOYMENTF_C == 12
    EMPLOYMENTF_C == 10 ~ 5 # unemployment
  )) %>%
  mutate(EMPLOYMENTF_C = as.factor(EMPLOYMENTF_C)) %>%
  mutate(EMPLOYMENTF = as.factor(EMPLOYMENTF)) %>%
  mutate(EMPLOY1 = as.factor(EMPLOY1))
```

Injury characteristics

```
# injury year
longitudinal_complete <- longitudinal_complete %>%
  mutate(INJYEAR = as.numeric(INJYEAR))

# cause of TBI (CauseCat)
## combine the cause of injury categories into: vehicular, violence, sports, falls, othe
longitudinal_complete <- longitudinal_complete %>%
  mutate(Cause = as.factor(Cause)) %>%
  mutate(CauseCat = case_when(
    (Cause == 1 | Cause == 2 | Cause == 3 | Cause == 4 | Cause == 5) ~ 1, # vehicular
    Cause == 10 ~ 2, #gunshot wound
    Cause == 11 ~ 3, # assault with blunt instrument
    Cause == 12 ~ 4, # other violence
    (Cause == 13 | Cause == 14 | Cause == 15 | Cause == 16 | Cause == 17 | Cause == 18)
    (Cause == 19) ~ 6, # falls
    (Cause == 20 | Cause == 21 | Cause == 22) ~ 7, # other
    (Cause == 999) ~ NA, TRUE ~ NA
  )) %>%
  mutate(CauseCat = ifelse(CauseCat == 999 | is.na(CauseCat), NA, CauseCat)) %>%
  mutate(Aetiology = case_when(
    CauseCat == 2 | CauseCat == 3 | CauseCat == 4 ~ 1,
    CauseCat ==  1 | CauseCat ==  5 | CauseCat ==  6 | CauseCat ==  7 ~ 0,
    TRUE ~ NA)
  ) %>%
  mutate(CauseCat = as.factor(CauseCat)) %>%
  mutate(Aetiology = as.factor(Aetiology))

# GCS scores includ. sedation category (GCSCatSed)
## those with NA are assigned the value of "999 = Unknown"
## recode the GCS variable into a categorical variable based on the criteria: mild (scor
longitudinal_complete <- longitudinal_complete %>%
  mutate(GCSCatSed = case_when(
    GCS < 9 ~ 1, # severe
    (GCS >= 9 & GCS < 13) ~ 2, # moderate
    (GCS >= 13 & GCS < 77) ~ 3, # mild
    GCS == 77 ~ 77, # sedated
    GCS == 88 ~ 88, # intubated
    GCS == 999 ~ 999,
    TRUE ~ GCS
  )) %>%
  mutate(GCSCatSed = ifelse(GCSCatSed == 999 | is.na(GCSCatSed), NA, GCSCatSed)) %>%
  mutate(GCSCatSed = as.factor(GCSCatSed)) %>%
  mutate(GCS = as.factor(GCS))

# spinal cord injury
longitudinal_complete <- longitudinal_complete %>%
  mutate(SCI_C = ifelse(SCI == 99 | is.na(SCI), NA, SCI)) %>%
  mutate(SCI_C = as.factor(SCI_C))

# hypertension
longitudinal_complete <- longitudinal_complete %>%
  mutate(CC_Hypertension = as.factor(CC_Hypertension)) %>%
  mutate(CC_Hypertension_C = ifelse(CC_Hypertension == 99 | is.na(CC_Hypertension), NA,
  mutate(CC_Hypertension_C = as.factor(CC_Hypertension_C))
```

History / repeated TBI

```
# TBI history
longitudinal_complete <- longitudinal_complete %>%
  mutate(TBIInjuryF = as.factor(TBIInjuryF)) %>%
  mutate(TBIInjuryF_C = if_else(TBIInjuryF %in% c(66, 77, 99) | is.na(TBIInjuryF), NA, T
  mutate(TBIInjuryF_C = as.factor(TBIInjuryF_C))

# TBI repeated TBI
longitudinal_complete <- longitudinal_complete %>%
  mutate(TBIRepInjuryF = as.factor(TBIRepInjuryF)) %>%
  mutate(TBIRepInjuryF_C = if_else(TBIRepInjuryF %in% c(66, 77, 99) | is.na(TBIRepInjury
  mutate(TBIRepInjuryF_C = as.factor(TBIRepInjuryF_C))
```

Disability

```
# disability rating scale (DRSF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(DRSF_C = ifelse(CombinedDRSF %in% c(777, 999) | is.na(CombinedDRSF), NA, Combin

# FIM total (FIMTOTF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(FIMTOTF_C = ifelse(FIMTOTF == 9999 | is.na(FIMTOTF), NA, FIMTOTF)) # a cleaned
# FIM cognitive (FIMCOGF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(FIMCOGF_C = ifelse(FIMCOGF == 9999 | is.na(FIMCOGF), NA, FIMCOGF)) # a cleaned

# GOS-E (GOSTotalF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(GOSTotalF_C = ifelse(GOSTotalF %in% c(66, 99) | is.na(GOSTotalF), NA, GOSTotalF
```

Cause of death

```
# age at death
longitudinal_complete <- longitudinal_complete %>%
  mutate(
    DAYStoDEATHF_C = ifelse(DAYStoDEATHF == 999999, NA, DAYStoDEATHF),
    AgeDeath = case_when(
      is.na(DAYStoDEATHF_C) ~ NA,
      TRUE ~ (DAYStoDEATHF_C/365.25)))
## imputed age at death when the DAYStoDEATHF variable is missing
longitudinal_complete <- longitudinal_complete %>%
  group_by(Mod1Id) %>%
  mutate(
    AgeNoPHI0 = AGENoPHIF_C[FollowUpPeriod == 0][1],
    AgeDeath_C = case_when(
      !is.na(AgeDeath) ~ AgeDeath,
      !is.na(DEATHYEAR) & is.na(AgeDeath) ~ (AgeNoPHI0 + (DEATHYEAR - INJYEAR))
    )
  ) %>%
  ungroup() %>%
  select(-AgeNoPHI0) %>%
  mutate(AgeDeath_C = as.numeric(AgeDeath_C))
## still cannot compute an age at death for those without an age at baseline or DAYStoDE

## death by assault
longitudinal_complete <- longitudinal_complete %>%
  mutate(DeathAssault = case_when(
    DeathECodeF %in% c("E9060", "E960.0", "E960.1", "E961", "E962", "E962.0", "E962.1",
    TRUE ~ 0
  ),
  DeathAssault = as.factor(DeathAssault))

## death by suicide
longitudinal_complete <- longitudinal_complete %>%
  mutate(DeathSuicide = case_when(
    DeathECodeF %in% c("E950", "E950.0", "E950.1", "E950.2", "E950.3", "E950.4", "E950.5
    TRUE ~ 0
  ),
  DeathSuicide = as.factor(DeathSuicide))

## death by overdose
longitudinal_complete <- longitudinal_complete %>%
  mutate(DeathOverdose = case_when(
    DeathCause1F %in% c("291", "291.0", "291.1", "291.2", "291.3", "291.4", "291.5", "29
    DeathCause2F %in% c("291", "291.0", "291.1", "291.2", "291.3", "291.4", "291.5", "29
    TRUE ~ 0
  ),
  DeathOverdose = as.factor(DeathOverdose))
```

Psychiatric illness and indices

```
# pre-TBI history of psychiatric hospitalisation (PsyHosp)
longitudinal_complete <- longitudinal_complete %>%
  mutate(PsyHosp = as.factor(PsyHosp)) %>%
  mutate(PsyHosp_C = ifelse(PsyHosp %in% c(66, 77, 88, 99) | is.na(PsyHosp), NA, PsyHosp
  mutate(PsyHosp_C = as.factor(PsyHosp_C))

# pre-TBI history of mental health treatment (MntlEver)
longitudinal_complete <- longitudinal_complete %>%
  mutate(MntlEver = as.factor(MntlEver)) %>%
  mutate(MntlEver_C = ifelse(MntlEver %in% c(66, 77, 99) | is.na(MntlEver), NA, MntlEver
  mutate(MntlEver_C = as.factor(MntlEver_C))

# suicide attempt (Suicide)
longitudinal_complete <- longitudinal_complete %>%
  mutate(SuicideF = as.factor(SuicideF)) %>%
  mutate(SuicideEver = ifelse(SuicideF %in% c(66, 77, 88, 99) | is.na(SuicideF), NA, Sui
  mutate(
    SuicidePrior = ifelse(SuicidePrior %in% c(66, 77, 99) | is.na(SuicidePrior), NA, Sui
    SuicideAttempt = case_when(
      is.na(SuicidePrior) ~ NA,
      SuicidePrior == 88 ~ 1, # never attempted suicide
      SuicidePrior == 0 ~ 2, # yes, but not in the last year
      SuicidePrior == 1 ~ 3 # yes, in the last year
  )) %>%
  mutate(SuicideEver = as.factor(SuicideEver)) %>%
  mutate(SuicideAttempt = as.factor(SuicideAttempt))

# drinking category (DRINKCatF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(DRINKCatF = as.factor(DRINKCatF)) %>%
  mutate(DRINKCatF_C = ifelse(DRINKCatF == 99 | is.na(DRINKCatF), NA, DRINKCatF)) %>%
  mutate(DRINKCatF_C = as.factor(DRINKCatF_C))

# problematic alcohol use
longitudinal_complete <- longitudinal_complete %>%
  mutate(PROBLEMUseF = as.factor(PROBLEMUseF)) %>%
  mutate(PROBLEMUseF_C = ifelse(PROBLEMUseF %in% c(77, 99) | is.na(PROBLEMUseF), NA, PRO
  mutate(PROBLEMUseF_C = as.factor(PROBLEMUseF_C))

# illicit drug use (DrugsF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(DrugsF = as.factor(DrugsF)) %>%
  mutate(DrugsF_C = ifelse(DrugsF %in% c(66, 77, 88, 99) | is.na(DrugsF), NA, DrugsF)) %
  mutate(DrugsF_C = as.factor(DrugsF_C))

# depression (PHQ9TOTF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(PHQ9TOTF_C= ifelse(PHQ9TOTF %in% c(666, 888, 999), NA, PHQ9TOTF))
longitudinal_complete <- longitudinal_complete %>%
  mutate(
    PHQPleasureF_C = if_else(PHQPleasureF %in% c(66, 81, 82, 99) | is.na(PHQPleasureF),
    PHQDownF_C = if_else(PHQDownF %in% c(66, 81, 82, 99) | is.na(PHQDownF), NA, PHQDownF
    PHQ2Count = PHQPleasureF_C + PHQDownF_C,
    gate_PHQ = case_when(
      PHQ2Count == 0 ~ 1,
      PHQ2Count > 0 ~ 2,
      TRUE ~ NA
    ),
    gate_PHQ = as.factor(gate_PHQ))

# anxiety (GAD7TOTF)
longitudinal_complete <- longitudinal_complete %>%
  mutate(GAD7TOTF_C= ifelse(GAD7TOTF %in% c(888, 999), NA, GAD7TOTF))
longitudinal_complete <- longitudinal_complete %>%
  mutate(
    GADNervousF_C = if_else(GADNervousF %in% c(66, 81, 82, 99) | is.na(GADNervousF), NA,
    GADCntrlWryF_C = if_else(GADCntrlWryF %in% c(66, 81, 82, 99) | is.na(GADCntrlWryF),
    GAD2Count = GADNervousF_C + GADCntrlWryF_C,
    gate_GAD = case_when(
      GAD2Count == 0 ~ 1,
      GAD2Count > 0 ~ 2,
      TRUE ~ NA
    ),
    gate_GAD = as.factor(gate_GAD))
```

Streamline

Streamline the dataset by selecting the "cleaned" version of each variable (ie, `Variable_C`).

```
longitudinal_clean <- longitudinal_complete %>%
  select(Mod1Id, Mod2Id, FollowUpPeriod, AGENoPHIF_C, SexF_C, INJYEAR, AgeDeath_C, DEATH

# for analyses looking at baseline characteristics restrict the sample to those with For
baseline_clean <- longitudinal_clean %>%
  filter(FollowUpPeriod == 0)
```

Codebook

Create a codebook for all variables using the package `labelled()`.

```
# Label your variables
var_label(longitudinal_clean) <- list(
  Mod1Id = "Unique identifier for rows in Form 1",
  Mod2Id = "Unique identifier for rows in Form 2",
  FollowUpPeriod = "Follow up period, ie, data collection wave following index TBI",
  AGENoPHIF_C = "Age at injury (unknown values were imputed based on age in Form 1)",
  SexF_C = "Sex. If transgender, current sex is recorded",
  INJYEAR = "Year of injury",
  AgeDeath_C = "Age at death (if age is unknown an estimated age at death based on death
  DEATHYEAR = "Year of death",
  DAYStoDEATHF_C = "Days from birth to death",
  DeathCause1F = "Primary cause of death, defined using ICD codes",
  DeathCause2F = "Secondary cause of death, defined using ICD codes",
```

```r
  DeathEcodeF = "External cause of death, defined using ICD codes",
  DeathAssault = "Binary variable indicating death by assault",
  DeathSuicide = "Binary variable indicating death by suicide",
  DeathOverdose = "Binary variable indicating that the primary or secondary cause of dea
  EduYears10_C = "Highest grade of school completed",
  EDUCATION_C = "Calculated educational attainment",
  EarnF_C = "Annual income",
  HHIncomeF_C = "Annual household income",
  RaceCat = "Racial category",
  Race1 = "Race where Pacific Islander and Native American are collapsed into one catego
  EthnicityF_C = "Self-reported ethnicity. Are you of Hispanic, Latino or Spanish origin
  CountryBirthF_C = "Country of birth",
  RURALF = "Urbanicity",
  EMPLOYMENTF_C = "Employment status",
  EMPLOY1 = "Calculated employment status with smaller categories",
  CauseCat = "Cause of injury",
  Aetiology = "Cause of injury dichotimised to violent or nonviolent",
  TBIInjuryF_C = "Is ther any head or neck injury reported",
  TBIRepInjuryF_C = "Have you ever had a period of time in which you experienced multipl
  GCSCatSed = "Glasgow Coma Score severity score",
  SCI_C = "Spinal cord injury",
  CC_Hypertension_C = "Intracranial hypertension",
  DRSF_C = "Disability Rating Scale score",
  FIMTOTF_C = "FIM total score",
  FIMCOGF_C = "FIM cognitive scale score",
  GOSTotalF_C = "Glasgow Outcome Scale total score",
  SpEd_C = "In school, were you ever classified as a special education student",
  PsyHosp_C = "Have you ever been hospitalised for a psychiatric problem?",
  MntlEver_C = "Have you ever recievied treatment for any mental health problem",
  SuicideEver = "Have you attempted suicide (lifetime)?",
  SuicideAttempt = "Have you attempted suicide, and if so did you attempt in the last ye
  DRINKCatF_C = "Alcohol consumption level",
  PROBLEMUseF = "Substance problem use",
  DrugsF_C = "Did you use any illict or non-prescription drugs in the last year?",
  gate_PHQ = "Did you endorse either the pleasure of down items on the PHQ?",
  PHQ9TOTF_C = "Patient Health Questionnaire (depression measure) total score",
  gate_GAD = "Did you endorse either the pleasure of down items on the GAD?",
  GAD7TOTF_C = "Generalised Anxiety Disorder (anxiety measure) total score"
)

codebook <- longitudinal_clean %>%
  look_for(details = "none")

datatable(codebook,
  options = list(
    scrollX = TRUE,              # enables horizontal scroll
    autoWidth = TRUE,            # adjusts column widths
    pageLength = 10              # optional: rows per page
  ),
  class = 'display nowrap stripe'
)
```

Show 10 entries

Search:

	pos	variable	label
1	1	Mod1Id	Unique identifier for rows in Form 1
2	2	Mod2Id	Unique identifier for rows in Form 2
3	3	FollowUpPeriod	Follow up period, ie, data collection wave following index TBI
4	4	AGENoPHIF_C	Age at injury (unknown values were imputed based on age in Form
5	5	SexF_C	Sex. If transgender, current sex is recorded
6	6	INJYEAR	Year of injury
7	7	AgeDeath_C	Age at death (if age is unknown an estimated age at death based or
8	8	DEATHYEAR	Year of death
9	9	DAYStoDEATHF_C	Days from birth to death
10	10	DeathCause1F	Primary cause of death, defined using ICD codes

Showing 1 to 10 of 52 entries

Previous 1 2 3 4 5 6 Next

Inclusion and exclusion criteria

Most variables of interest are available in the dataset from October 2012. In order to reduce structural missingness, the sample was restricted to those enrolled in TBIMS between 2013 and 2023, reflecting the period when all variables of interest were available in the TBIMS NDB.

```r
data_risk <- baseline_clean %>%                                                       ⑦
  filter(INJYEAR >= 2013 & INJYEAR <= 2023) %>%                                       ⑧
  select(-Mod2Id, -DAYStoDEATHF_C, -HHIncomeF_C, -gate_PHQ, -PHQ9TOTF_C, -gate_GAD, - ⑨
```

⑦ As this study is a cross-sectional study I have filtered for the baseline/Form 1 data only.

⑧ Restrict the sample to those enrolled in TBIMS between 2013 and 2023.

⑨ Variables that only appear in Form 2, and thus are not available at baseline, are removed.

Relative to their younger counterparts, older adults more often sustain TBI from falls and less often from vehicular or sports-related incidents and violence. We chose to restrict the analysis to those aged under 55 due to distinct injury mechanism patterns.

```r
data_risk <- data_risk %>%
  filter(AGENoPHIF_C < 55)
```

Rename variables for ease of interpretation in results, now that the cohort is finalised.

```r
data_risk <- data_risk %>%
  select(Mod1Id, CauseCat, Aetiology, PsyHosp_C, MntlEver_C, SuicideEver, SuicideAttempt

data_risk %>%
  distinct(Mod1Id) %>%
  nrow()
```

```
[1] 5050
```

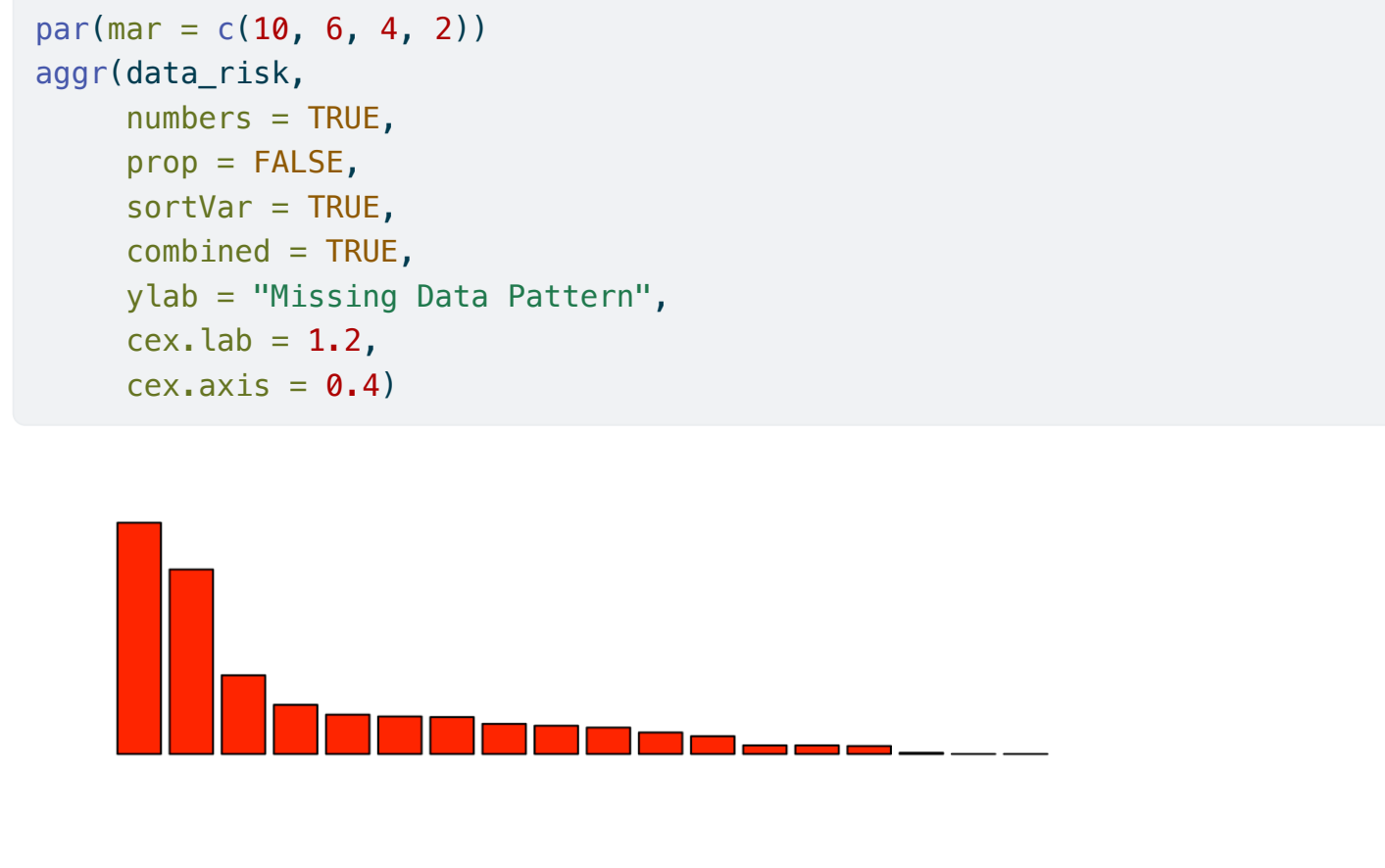
```r
data_risk <- data_risk %>%
  dplyr::rename(
    Age = AGENoPHIF_C,
    Sex = SexF_C,
    Race = race_ethnicity_C,
    Urbanicity = RURALF,
    Drugs = DrugsF_C,
    Suicide_Lifetime = SuicideEver,
    Suicide_Recent = SuicideAttempt,
    Special_Education = SpEd_C,
    Mental_Treatment = MntlEver_C,
    Psychiatric_Hospitalisation = PsyHosp_C,
    Education = EDUCATION_C,
    Employment = EMPLOY1,
    GCS = GCSCatSed,
    Cause = CauseCat,
    Aetiology = Aetiology,
    Alcohol_Use = DRINKCatF_C,
    BornUS = CountryBirthF_C
  )
```

```r
data_risk <- data_risk %>%
  mutate(Sex = factor(Sex, labels = c("Female", "Male")),
         Race = factor(Race, labels = c("White", "Black or AA", "Asian / NHPI", "Hispani
         BornUS = factor(BornUS, labels = c("Born in the US", "Born outside the US")),
         Urbanicity = factor(Urbanicity, labels = c("Rural", "Urban", "Suburban")),
         Special_Education = factor(Special_Education, labels = c("No", "Yes")),
         Employment = factor(Employment, labels = c("Competitively employed", "Student",
         Education = factor(Education, labels = c("In school", "HS/GED", "Some college",
         Cause = factor(Cause, labels = c("Vehicular", "Gunshot wounds", "Assault with b
         Aetiology = factor(Aetiology, labels = c("Nonviolent", "Violent")),
         Mental_Treatment = factor(Mental_Treatment, labels = c("No", "Yes")),
         Psychiatric_Hospitalisation = factor(Psychiatric_Hospitalisation, labels = c("N
         Alcohol_Use = factor(Alcohol_Use, labels = c("Abstaining", "Light", "Moderate",
         Suicide_Lifetime = factor(Suicide_Lifetime, labels = c("No", "Yes")),
         Suicide_Recent = factor(Suicide_Recent, labels = c("No", "Yes but not in the la
         Drugs = factor(Drugs, labels = c("No", "Yes")),
         GCS = factor(GCS, labels = c("Severe", "Moderate", "Mild", "Sedated", "Intubate
         )
```

Missing data

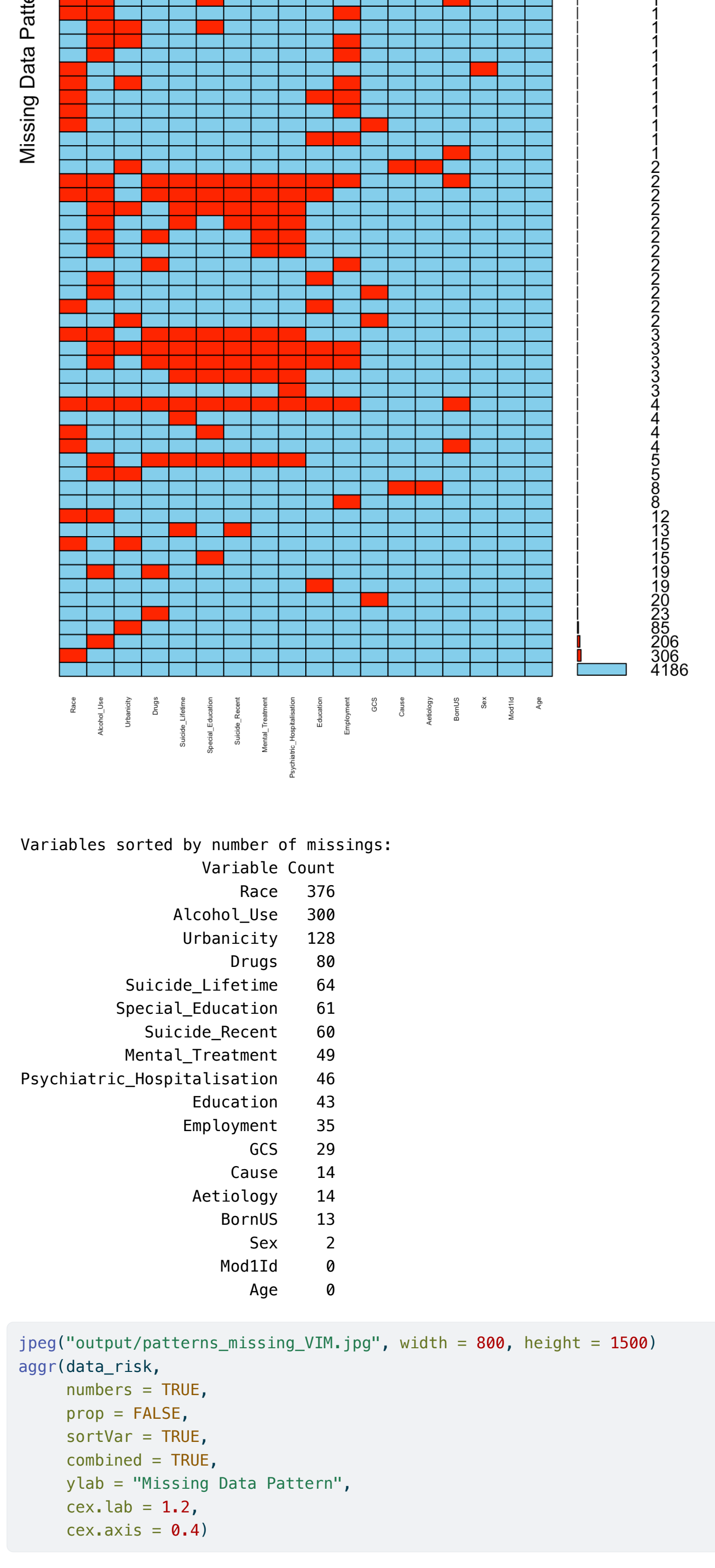
Visualise missingness

```r
data_risk %>%
  select(-Mod1Id) %>%
  gg_miss_var(show_pct = TRUE) + labs(title = "Proportion of missingness")
```



```r
jpeg("output/naniar_missing.jpg", width = 800, height = 600)
data_risk %>%
  select(-Mod1Id) %>%
  gg_miss_var(show_pct = TRUE) + labs(title = "Proportion of missingness")
dev.off()
```

```r
par(mar = c(10, 6, 4, 2))
aggr(data_risk,
     numbers = TRUE,
     prop = FALSE,
     sortVar = TRUE,
     combined = TRUE,
     ylab = "Missing Data Pattern",
     cex.lab = 1.2,
     cex.axis = 0.4)
```



```
 Variables sorted by number of missings:
                    Variable Count
                        Race   376
                 Alcohol_Use   300
                  Urbanicity   128
                       Drugs    80
            Suicide_Lifetime    64
           Special_Education    61
              Suicide_Recent    60
            Mental_Treatment    49
 Psychiatric_Hospitalisation    46
                   Education    43
                  Employment    35
                         GCS    29
                       Cause    14
                   Aetiology    14
                      BornUS    13
                         Sex     2
                      Mod1Id     0
                         Age     0
```

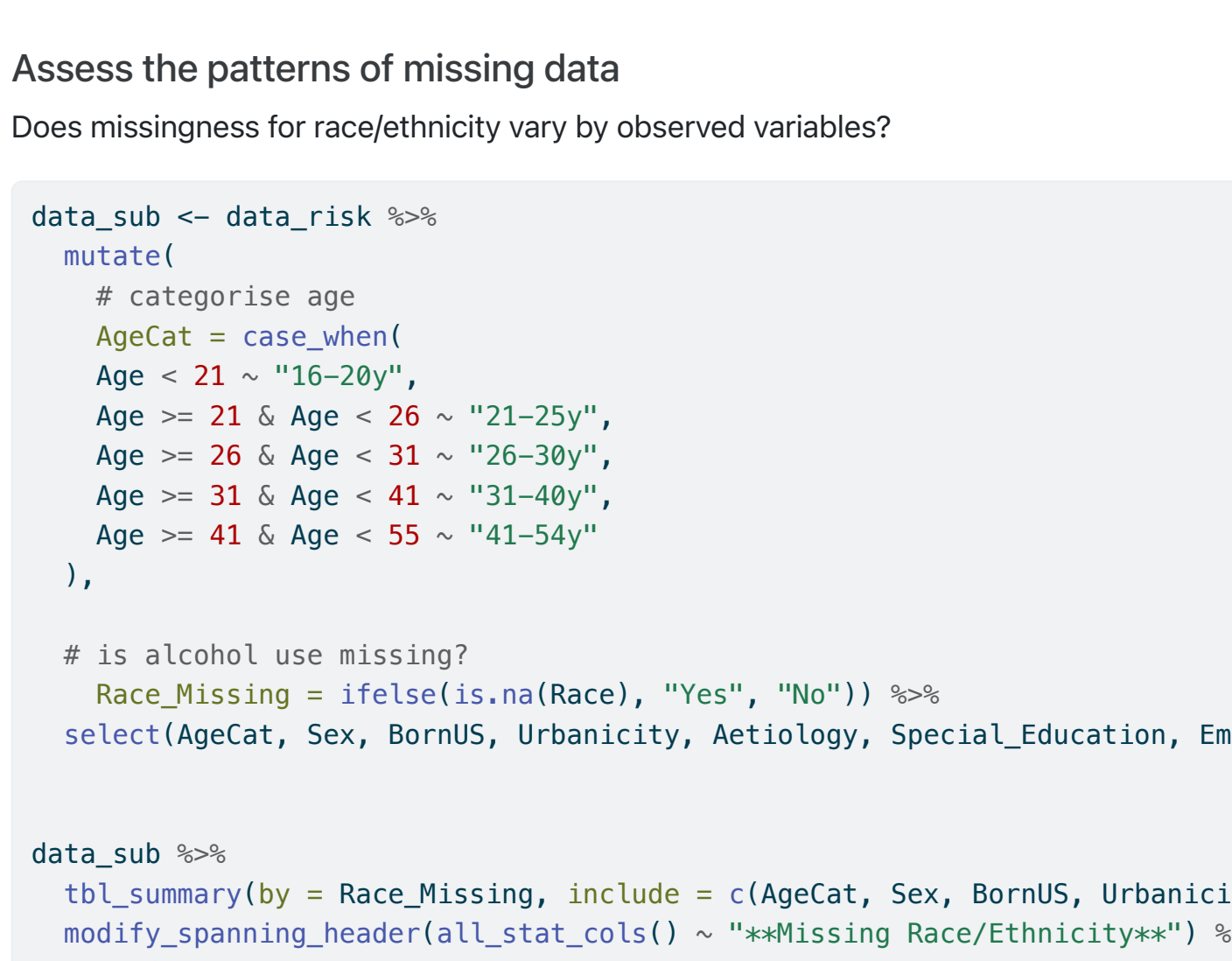
```r
jpeg("output/patterns_missing_VIM.jpg", width = 800, height = 1500)
aggr(data_risk,
     numbers = TRUE,
     prop = FALSE,
     sortVar = TRUE,
     combined = TRUE,
     ylab = "Missing Data Pattern",
     cex.lab = 1.2,
     cex.axis = 0.4)
```

```
 Variables sorted by number of missings:
                    Variable Count
                        Race   376
                 Alcohol_Use   300
                  Urbanicity   128
                       Drugs    80
            Suicide_Lifetime    64
           Special_Education    61
              Suicide_Recent    60
            Mental_Treatment    49
 Psychiatric_Hospitalisation    46
                   Education    43
                  Employment    35
                         GCS    29
                       Cause    14
                   Aetiology    14
                      BornUS    13
                         Sex     2
                      Mod1Id     0
                         Age     0
```

```r
dev.off()
```

```
quartz_off_screen
                2
```

```r
par(mar = c(10, 6, 4, 2))
aggr(data_risk %>%
       select(Alcohol_Use, Drugs, Psychiatric_Hospitalisation, Mental_Treatment, Suicide
     numbers = TRUE,
     prop = FALSE,
     sortVar = TRUE,
     combined = FALSE,
     cex.lab = 1.2,
     cex.axis = 0.4)
```



```
 Variables sorted by number of missings:
                    Variable Count
                 Alcohol_Use   300
                       Drugs    80
            Suicide_Lifetime    64
              Suicide_Recent    60
            Mental_Treatment    49
 Psychiatric_Hospitalisation    46
```

Assess the patterns of missing data

Does missingness for race/ethnicity vary by observed variables?

```r
data_sub <- data_risk %>%
  mutate(
    # categorise age
    AgeCat = case_when(
    Age < 21 ~ "16-20y",
    Age >= 21 & Age < 26 ~ "21-25y",
    Age >= 26 & Age < 31 ~ "26-30y",
    Age >= 31 & Age < 41 ~ "31-40y",
    Age >= 41 & Age < 55 ~ "41-54y"
  ),

  # is alcohol use missing?
    Race_Missing = ifelse(is.na(Race), "Yes", "No")) %>%
  select(AgeCat, Sex, BornUS, Urbanicity, Aetiology, Special_Education, Employment, Educ


data_sub %>%
  tbl_summary(by = Race_Missing, include = c(AgeCat, Sex, BornUS, Urbanicity, Special_Ed
  modify_spanning_header(all_stat_cols() ~ "**Missing Race/Ethnicity**") %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
test.args = all_categorical() ~ list(simulate.p.value = TRUE))
```

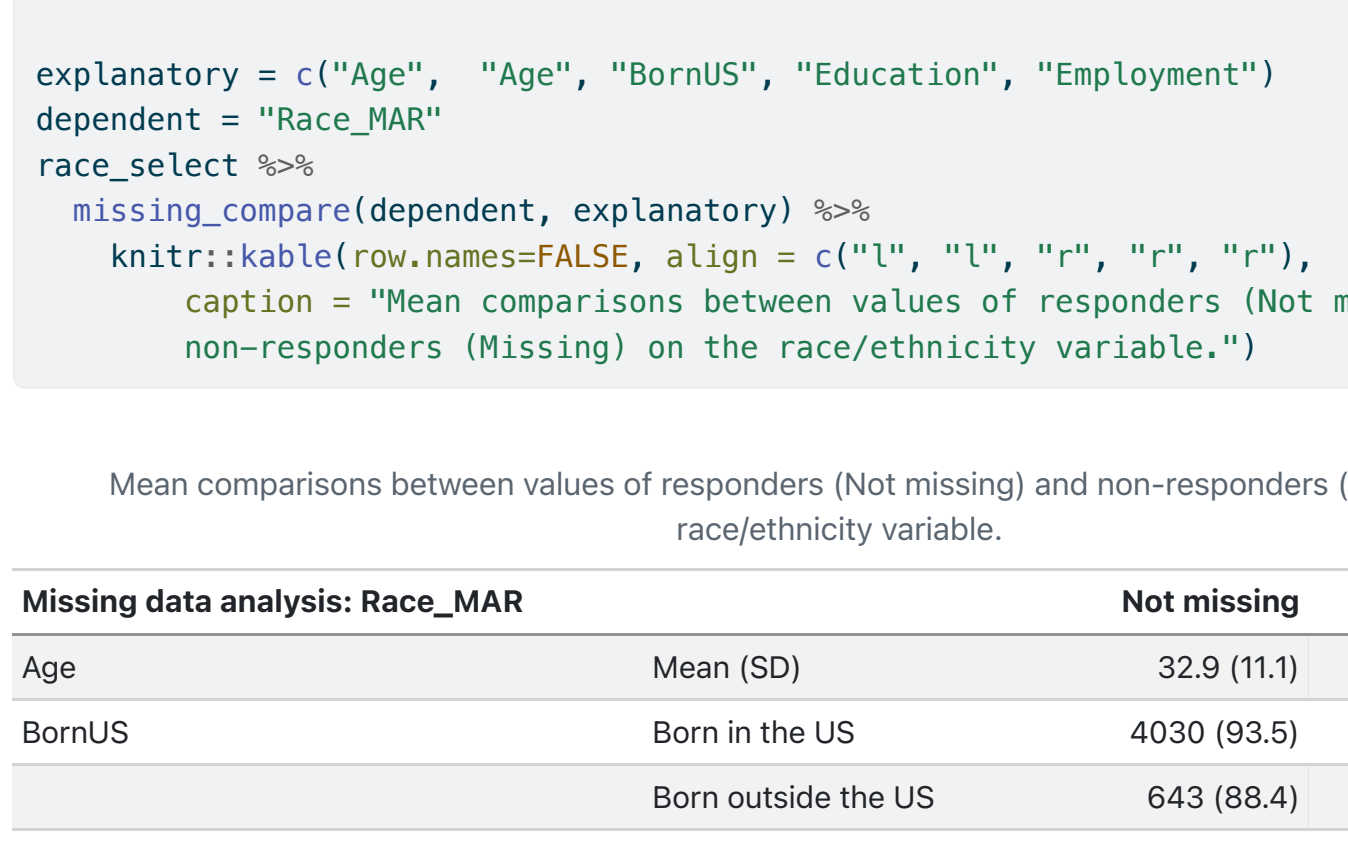
	Missing Race/Ethnicity		
Characteristic	**No** N = 4,674[1]	**Yes** N = 376[1]	**p-value**[2]
AgeCat			0.021
16-20y	681 (15%)	34 (9.0%)	
21-25y	904 (19%)	73 (19%)	
26-30y	708 (15%)	57 (15%)	
31-40y	1,049 (22%)	83 (22%)	
41-54y	1,332 (28%)	129 (34%)	
Sex			0.10
Female	1,087 (23%)	73 (19%)	
Male	3,586 (77%)	302 (81%)	
Unknown	1	1	
BornUS			<0.001
Born in the US	4,030 (86%)	280 (77%)	
Born outside the US	643 (14%)	84 (23%)	
Unknown	1	12	
Urbanicity			0.010
Rural	1,131 (25%)	79 (22%)	
Urban	2,112 (46%)	191 (54%)	
Suburban	1,327 (29%)	82 (23%)	
Unknown	104	24	
Special_Education	474 (10%)	48 (14%)	0.039
Unknown	37	24	
Employment			0.003
Competitively employed	3,429 (74%)	253 (70%)	
Student	349 (7.5%)	16 (4.4%)	
Retired	181 (3.9%)	15 (4.1%)	
Other employment	116 (2.5%)	10 (2.8%)	
Unemployed	578 (12%)	68 (19%)	
Unknown	21	14	
Education			0.010
In school	922 (20%)	98 (27%)	
HS/GED	1,800 (39%)	142 (39%)	
Some college	939 (20%)	51 (14%)	
Associates degree	299 (6.4%)	19 (5.2%)	
Bachelors degree	509 (11%)	40 (11%)	
Graduate degree	176 (3.8%)	12 (3.3%)	
Unknown	29	14	

[1] n (%)

[2] Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

```r
tf <- "output/gtsummary-race-missing.docx"

data_sub %>%
  tbl_summary(by = Race_Missing, include = c(AgeCat, Sex, BornUS, Urbanicity, Special_Ed
  modify_spanning_header(all_stat_cols() ~ "**Missing Race/Ethnicity**") %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  as_gt() %>%
  gt::gtsave(filename = tf)
```

```r
age.race <- data_risk[, c("Race", "Age")]
marginplot(age.race)
```



```r
jpeg("output/age_race_marginplot.jpg", width = 800, height = 600)
marginplot(age.race)
dev.off()
```

```
quartz_off_screen
                2
```

```r
race_select <- data.frame(data_risk[,c("Race", "Age", "BornUS", "Education", "Employment

# copy variable with missing data and rename
race_select$Race_MAR <- race_select$Race

explanatory = c("Age",  "Age", "BornUS", "Education", "Employment")
dependent = "Race_MAR"
race_select %>%
  missing_compare(dependent, explanatory) %>%
    knitr::kable(row.names=FALSE, align = c("l", "l", "r", "r", "r"),
        caption = "Mean comparisons between values of responders (Not missing) and
        non-responders (Missing) on the race/ethnicity variable.")
```

Mean comparisons between values of responders (Not missing) and non-responders (Missing) on the race/ethnicity variable.

Missing data analysis: Race_MAR		Not missing	Missing	p
Age	Mean (SD)	32.9 (11.1)	34.4 (11.1)	0.011
BornUS	Born in the US	4030 (93.5)	280 (6.5)	<0.001
	Born outside the US	643 (88.4)	84 (11.6)	
Education	In school	922 (90.4)	98 (9.6)	0.007
	HS/GED	1800 (92.7)	142 (7.3)	
	Some college	939 (94.8)	51 (5.2)	
	Associates degree	299 (94.0)	19 (6.0)	
	Bachelors degree	509 (92.7)	40 (7.3)	
	Graduate degree	176 (93.6)	12 (6.4)	
Employment	Competitively employed	3429 (93.1)	253 (6.9)	0.003
	Student	349 (95.6)	16 (4.4)	
	Retired	181 (92.3)	15 (7.7)	
	Other employment	116 (92.1)	10 (7.9)	
	Unemployed	578 (89.5)	68 (10.5)	

Does missingness for alcohol use vary by observed variables?

```r
data_sub <- data_risk %>%
  mutate(
    # categorise age
    AgeCat = case_when(
    Age < 21 ~ "16-20y",
    Age >= 21 & Age < 26 ~ "21-25y",
    Age >= 26 & Age < 31 ~ "26-30y",
    Age >= 31 & Age < 41 ~ "31-40y",
    Age >= 41 & Age < 55 ~ "41-54y"
  ),

  # is alcohol use missing?
    Alcohol_Missing = ifelse(is.na(Alcohol_Use), "Yes", "No")) %>%
  select(AgeCat, Sex, Race, BornUS, Urbanicity, Aetiology, Special_Education, Employment


data_sub %>%
  tbl_summary(by = Alcohol_Missing, include = c(AgeCat, Sex, Race, BornUS, Urbanicity, S
  modify_spanning_header(all_stat_cols() ~ "**Missing Alcohol Consumption**") %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
test.args = all_categorical() ~ list(simulate.p.value = TRUE))
```

	Missing Alcohol Consumption		
Characteristic	**No** N = 4,750[1]	**Yes** N = 300[1]	**p-value**[2]
AgeCat			0.3
16-20y	673 (14%)	42 (14%)	
21-25y	913 (19%)	64 (21%)	
26-30y	714 (15%)	51 (17%)	
31-40y	1,079 (23%)	53 (18%)	
41-54y	1,371 (29%)	90 (30%)	
Sex			0.081
Female	1,104 (23%)	56 (19%)	
Male	3,644 (77%)	244 (81%)	
Unknown	2	0	
Race			0.8
White	2,395 (54%)	157 (59%)	
Black or AA	834 (19%)	45 (17%)	
Asian / NHPI	123 (2.8%)	5 (1.9%)	
Hispanic or Latino	905 (21%)	49 (18%)	
AIAN	16 (0.4%)	1 (0.4%)	
Non-Hispanic Other	25 (0.6%)	1 (0.4%)	
More than one race	111 (2.5%)	7 (2.6%)	
Unknown	341	35	
BornUS			0.5
Born in the US	4,064 (86%)	246 (84%)	
Born outside the US	681 (14%)	46 (16%)	
Unknown	5	8	
Urbanicity			>0.9
Rural	1,139 (25%)	71 (25%)	
Urban	2,171 (47%)	132 (47%)	
Suburban	1,332 (29%)	77 (28%)	
Unknown	108	20	
Special_Education	486 (10%)	36 (14%)	0.10
Unknown	26	35	
Employment			0.4
Competitively employed	3,483 (74%)	199 (71%)	
Student	349 (7.4%)	16 (5.7%)	
Retired	183 (3.9%)	13 (4.6%)	
Other employment	118 (2.5%)	8 (2.8%)	
Unemployed	601 (13%)	45 (16%)	
Unknown	16	19	
Education			0.018
In school	963 (20%)	57 (20%)	
HS/GED	1,807 (38%)	135 (48%)	
Some college	944 (20%)	46 (16%)	

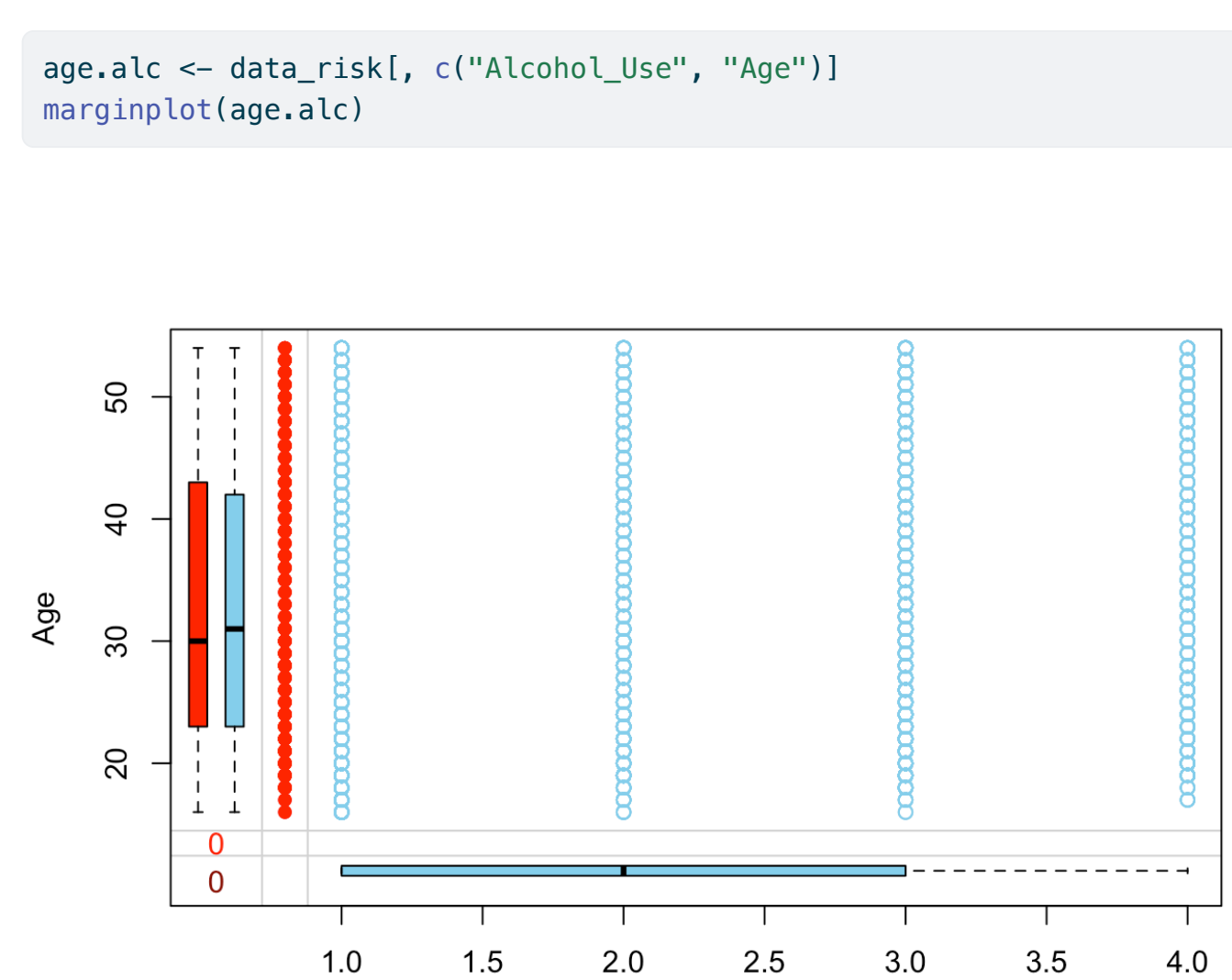
Associates degree	305 (6.5%)	13 (4.6%)	
Bachelors degree	524 (11%)	25 (8.9%)	
Graduate degree	183 (3.9%)	5 (1.8%)	
Unknown	24	19	
Mental_Treatment	1,123 (24%)	65 (25%)	0.7
Unknown	10	39	
Drugs	1,271 (27%)	82 (33%)	0.039
Unknown	29	51	

[1] n (%)

[2] Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

```
tf2 <- "output/gtsummary-alcohol-missing.docx"

data_sub %>%
  tbl_summary(by = Alcohol_Missing, include = c(AgeCat, Sex, Race, BornUS, Urbanicity, S
  modify_spanning_header(all_stat_cols() ~ "**Missing Alcohol Consumption**") %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  as_gt() %>%
  gt::gtsave(filename = tf2)
```

```
age.alc <- data_risk[, c("Alcohol_Use", "Age")]
marginplot(age.alc)
```



```
jpeg("output/age_alcohol_marginplot.jpg", width = 800, height = 600)
marginplot(age.alc)
dev.off()
```

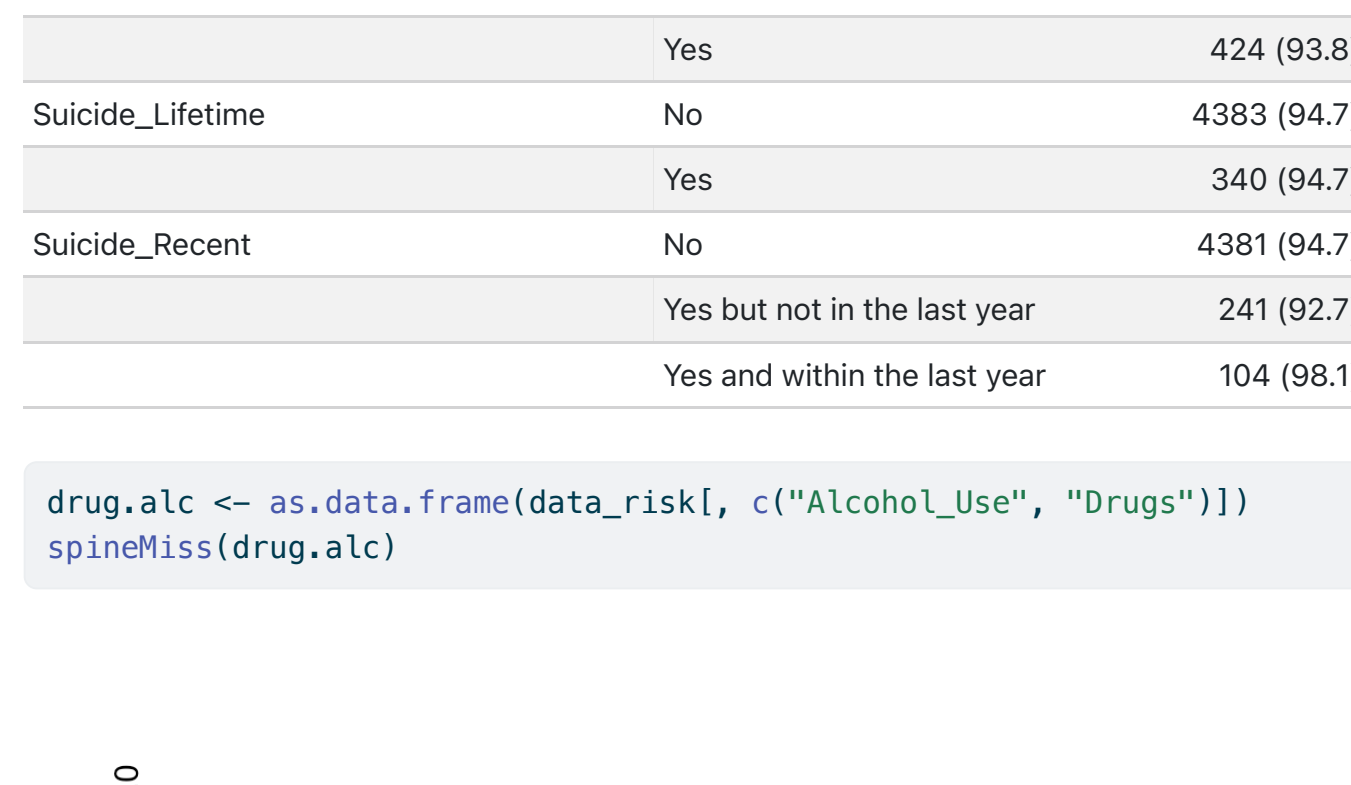
```
quartz_off_screen
                2
```

```
alc_select <- data.frame(data_risk[,c("Alcohol_Use", "Age", "Drugs", "Mental_Treatment",

# copy variable with missing data and rename
alc_select$Alcohol_MAR <- alc_select$Alcohol_Use

explanatory = c("Age", "Drugs", "Mental_Treatment", "Psychiatric_Hospitalisation", "Suic
dependent = "Alcohol_MAR"
alc_select %>%
  missing_compare(dependent, explanatory) %>%
    knitr::kable(row.names=FALSE, align = c("l", "l", "r", "r", "r"),
      caption = "Mean comparisons between values of responders (Not missing) and
      non-responders (Missing) on the alcohol use variable.")
```

Mean comparisons between values of responders (Not missing) and non-responders (Missing) on the alcohol use variable.

Missing data analysis: Alcohol_MAR		Not missing	Missing	p
Age	Mean (SD)	33.0 (11.1)	32.9 (11.4)	0.856
Drugs	No	3450 (95.4)	167 (4.6)	0.045
	Yes	1271 (93.9)	82 (6.1)	
Mental_Treatment	No	3617 (94.9)	196 (5.1)	0.709
	Yes	1123 (94.5)	65 (5.5)	
Psychiatric_Hospitalisation	No	4317 (94.8)	235 (5.2)	0.408
	Yes	424 (93.8)	28 (6.2)	
Suicide_Lifetime	No	4383 (94.7)	244 (5.3)	1.000
	Yes	340 (94.7)	19 (5.3)	
Suicide_Recent	No	4381 (94.7)	243 (5.3)	0.102
	Yes but not in the last year	241 (92.7)	19 (7.3)	
	Yes and within the last year	104 (98.1)	2 (1.9)	

```
drug.alc <- as.data.frame(data_risk[, c("Alcohol_Use", "Drugs")])
spineMiss(drug.alc)
```



```
Click in in the left margin to switch to the previous variable or in the right margin to
To regain use of the VIM GUI and the R console, click anywhere else in the graphics windo
```

```
jpeg("output/drug_alcohol_spineplot.jpg", width = 800, height = 600)
spineMiss(drug.alc)
```

```
Click in in the left margin to switch to the previous variable or in the right margin to
To regain use of the VIM GUI and the R console, click anywhere else in the graphics windo
```

```
dev.off()
```

```
quartz_off_screen
                2
```

Multiple imputation

Fitting the imputed model

Include all variables that will be used in the analysis in the same form they will be used in the model.

```
data_risk <- data_risk %>%
  select(-Aetiology) # <5> I don't include `Aetiology` in the imputation model, as `Aeti

imp.mids <- mice(data_risk,
                 seed   = 3, # ensure reproducibility
                 m      = 20, # when you are ready to fit your final imputation model,
                 print = F)
```

```
Warning: glm.fit: algorithm did not converge
Warning: glm.fit: algorithm did not converge
```

```
summary(imp.mids)
```

```
Class: mids
Number of multiple imputations:  20
Imputation methods:
                     Mod1Id                       Cause
                         ""                   "polyreg"
Psychiatric_Hospitalisation            Mental_Treatment
                   "logreg"                    "logreg"
           Suicide_Lifetime              Suicide_Recent
                   "logreg"                   "polyreg"
                      Drugs                 Alcohol_Use
                   "logreg"                   "polyreg"
                        Age                         Sex
                         ""                    "logreg"
                       Race                  Urbanicity
                  "polyreg"                   "polyreg"
                  Education           Special_Education
                  "polyreg"                    "logreg"
                 Employment                      BornUS
                  "polyreg"                    "logreg"
                        GCS
                  "polyreg"
PredictorMatrix:
                            Mod1Id Cause Psychiatric_Hospitalisation
Mod1Id                           0     1                           1
Cause                            1     0                           1
Psychiatric_Hospitalisation      1     1                           0
Mental_Treatment                 1     1                           1
Suicide_Lifetime                 1     1                           1
Suicide_Recent                   1     1                           1
                            Mental_Treatment Suicide_Lifetime Suicide_Recent
Mod1Id                                     1                1              1
Cause                                      1                1              1
Psychiatric_Hospitalisation                1                1              1
Mental_Treatment                           0                1              1
Suicide_Lifetime                           1                0              1
Suicide_Recent                             1                1              0
                            Drugs Alcohol_Use Age Sex Race Urbanicity Education
Mod1Id                          1           1   1   1    1          1         1
Cause                           1           1   1   1    1          1         1
Psychiatric_Hospitalisation     1           1   1   1    1          1         1
Mental_Treatment                1           1   1   1    1          1         1
Suicide_Lifetime                1           1   1   1    1          1         1
Suicide_Recent                  1           1   1   1    1          1         1
                            Special_Education Employment BornUS GCS
Mod1Id                                      1          1      1   1
Cause                                       1          1      1   1
Psychiatric_Hospitalisation                 1          1      1   1
Mental_Treatment                            1          1      1   1
Suicide_Lifetime                            1          1      1   1
Suicide_Recent                              1          1      1   1
```

Return the imputed dataset using `complete()`.

```
imp.data <- complete(imp.mids)  # return the first imputed dataset
```

Does the imputed dataset have the same dimensions as the original data?

```
dim(data_risk)
```

```
[1] 5050   17
```

```
dim(imp.data)
```

```
[1] 5050   17
```

Return all imputed datasets in "long" format (with stacking) including the original data

```
imp.data <- complete(imp.mids, "long", include = TRUE)
class(imp.data)
```

```
[1] "data.frame"
```

```
dim(imp.data)
```

```
[1] 106050     19
```

```
names(imp.data)

range(imp.data$.imp)
range(imp.data$.id)
```
④

④ A new variable called ".imp" indexes the imputations, where .imp = 0 corresponds to the original data and .imp = 1 to m corresponds to th imputed data. The new variable ".id" indexes the observations

```
 [1] "Mod1Id"                      "Cause"
 [3] "Psychiatric_Hospitalisation" "Mental_Treatment"
 [5] "Suicide_Lifetime"            "Suicide_Recent"
 [7] "Drugs"                       "Alcohol_Use"
 [9] "Age"                         "Sex"
[11] "Race"                        "Urbanicity"
[13] "Education"                   "Special_Education"
[15] "Employment"                  "BornUS"
[17] "GCS"                         ".imp"
[19] ".id"
[1]  0 20
[1]    1 5050
```

Dichotomise for the outcome (violent vs nonviolent TBI)

```
imp.data <- imp.data %>%
  mutate(
    Aetiology = case_when(
    Cause == "Vehicular" | Cause == "Sports" | Cause == "Falls" | Cause == "Other" ~ 0,
    Cause == "Gunshot wounds" | Cause == "Assault with blunt instrument" | Cause == "Oth
  ),
  Aetiology = factor(Aetiology, labels = c("Nonviolent", "Violent")))

imp.risk <- as.mids(imp.data, where = NULL, .imp = ".imp", .id = ".id")
```

Examine the imputed values

Create a table comparing the distribution of the variable between the complete data and the imputed data.

```
imp.data <- imp.data %>%
  mutate(imputed = .imp > 0,
         imputed = factor(imputed,
                          levels = c(FALSE, TRUE),
                          labels = c("Observed", "Imputed")))

prop_observed_imputed <- function(data, vars, imp_var) {

  data %>%
    select(all_of(c(vars, imp_var))) %>%
    tbl_summary(
      by = !!sym(imp_var),
      statistic = all_categorical() ~ "{p}%",
      percent = "column",
      missing = "no"
    ) %>%
    add_p() %>%
    bold_labels()
}

prop_observed_imputed(
  data   = imp.data,
  vars   = c("Sex", "Race", "BornUS", "Urbanicity", "Employment", "Education",
             "Special_Education", "GCS", "Cause", "Aetiology",
             "Psychiatric_Hospitalisation", "Mental_Treatment",
             "Suicide_Lifetime", "Suicide_Recent", "Alcohol_Use", "Drugs"),
  imp_var = "imputed"
)
```

Characteristic	Observed N = 5,050[1]	Imputed N = 101,000[1]	p-value[2]
Sex			>0.9
Female	23%	23%	
Male	77%	77%	
Race			>0.9
White	55%	54%	
Black or AA	19%	19%	
Asian / NHPI	2.7%	2.8%	
Hispanic or Latino	20%	21%	
AIAN	0.4%	0.4%	
Non-Hispanic Other	0.6%	0.6%	
More than one race	2.5%	2.5%	
BornUS			>0.9
Born in the US	86%	86%	
Born outside the US	14%	14%	
Urbanicity			>0.9
Rural	25%	24%	
Urban	47%	47%	
Suburban	29%	29%	
Employment			>0.9
Competitively employed	73%	73%	
Student	7.3%	7.3%	
Retired	3.9%	3.9%	
Other employment	2.5%	2.5%	
Unemployed	13%	13%	
Education			>0.9
In school	20%	20%	
HS/GED	39%	39%	
Some college	20%	20%	
Associates degree	6.4%	6.4%	
Bachelors degree	11%	11%	
Graduate degree	3.8%	3.8%	
Special_Education	10%	10%	>0.9
GCS			>0.9
Severe	37%	37%	
Moderate	11%	11%	
Mild	22%	22%	
Sedated	30%	29%	
Intubated	0.3%	0.3%	
Cause			>0.9
Vehicular	56%	56%	
Gunshot wounds	5.6%	5.6%	
Assault with blunt instrument	4.7%	4.7%	
Other violence	1.4%	1.4%	
Sports	2.5%	2.5%	
Falls	19%	19%	
Other	11%	11%	
Aetiology			>0.9
Nonviolent	88%	88%	
Violent	12%	12%	
Psychiatric_Hospitalisation	9.0%	9.0%	>0.9
Mental_Treatment	24%	24%	>0.9
Suicide_Lifetime	7.2%	7.3%	0.7
Suicide_Recent			>0.9
No	93%	93%	
Yes but not in the last year	5.2%	5.3%	
Yes and within the last year	2.1%	2.1%	
Alcohol_Use			>0.9
Abstaining	36%	36%	
Light	21%	21%	
Moderate	28%	28%	
Heavy	15%	15%	
Drugs	27%	27%	>0.9

[1]

[2] Pearson's Chi-squared test

Complete case analysis

How many rows would we lose if we required complete cases for these predictors?

```
data_complete <- data_risk

data_miss <- data_complete

n_total <- nrow(data_miss)
n_cc    <- data_miss %>%
  complete.cases() %>%
  sum()

cbind(
  total_rows    = n_total,
  complete_cases= n_cc,
  lost_rows     = n_total - n_cc,
  lost_pct      = round((n_total - n_cc) / n_total * 100, 1)
)
```

```
     total_rows complete_cases lost_rows lost_pct
[1,]       5050           4186       864     17.1
```

Descriptive statistics

After MI, compute descriptive statistics within each complete dataset and use Rubin's rules to **pool** statistics and compute their standard errors.

Use the `mi.mean.se.sd()` function (found in `Functions_rmph.R`) to calculate descriptive for the continuous variable `AGENoPHI`. The functions within `Functions_rmph.R` operate on a `mids` object.

```
# use the original 'mids' object imp.risk
mi.mean.se.sd(imp.risk, "Age")
```

```
        mean       se       sd
Age 32.96634 0.156804 11.14302
```

```
# frequencies and proportions
rbind(mi.n.p(imp.risk, "Aetiology"),
      mi.n.p(imp.risk, "Sex"),
      mi.n.p(imp.risk, "Race"),
      mi.n.p(imp.risk, "Urbanicity"),
      mi.n.p(imp.risk, "BornUS"),
      mi.n.p(imp.risk, "Education"),
      mi.n.p(imp.risk, "Special_Education"),
      mi.n.p(imp.risk, "Employment"),
      mi.n.p(imp.risk, "GCS"),
      mi.n.p(imp.risk, "Cause"),
      mi.n.p(imp.risk, "Mental_Treatment"),
      mi.n.p(imp.risk, "Psychiatric_Hospitalisation"),
      mi.n.p(imp.risk, "Suicide_Lifetime"),
      mi.n.p(imp.risk, "Alcohol_Use"),
      mi.n.p(imp.risk, "Drugs")
      )
```

```
                                         n           p
Aetiology: Nonviolent              4459.85 0.883138614
Aetiology: Violent                  590.15 0.116861386
Sex: Female                        1160.50 0.229801980
Sex: Male                          3889.50 0.770198020
Race: White                        2732.10 0.541009901
Race: Black or AA                   954.45 0.189000000
Race: Asian / NHPI                  139.80 0.027683168
Race: Hispanic or Latino           1049.80 0.207881188
Race: AIAN                           19.40 0.003841584
Race: Non-Hispanic Other             28.35 0.005613861
Race: More than one race            126.10 0.024970297
Urbanicity: Rural                  1233.20 0.244198020
Urbanicity: Urban                  2377.35 0.470762376
Urbanicity: Suburban               1439.45 0.285039604
BornUS: Born in the US             4320.40 0.855524752
BornUS: Born outside the US         729.60 0.144475248
Education: In school               1030.05 0.203970297
Education: HS/GED                  1957.50 0.387623762
Education: Some college             997.40 0.197504950
Education: Associates degree        321.05 0.063574257
Education: Bachelors degree         553.55 0.109613861
Education: Graduate degree          190.45 0.037712871
Special_Education: No              4521.10 0.895267327
Special_Education: Yes              528.90 0.104732673
Employment: Competitively employed 3703.95 0.733455446
Employment: Student                 367.75 0.072821782
Employment: Retired                 197.35 0.039079208
Employment: Other employment        127.40 0.025227723
Employment: Unemployed              653.55 0.129415842
GCS: Severe                        1856.70 0.367663366
GCS: Moderate                       571.65 0.113198020
GCS: Mild                          1119.50 0.221683168
GCS: Sedated                       1488.00 0.294653465
GCS: Intubated                       14.15 0.002801980
Cause: Vehicular                   2811.25 0.556683168
Cause: Gunshot wounds               283.60 0.056158416
Cause: Assault with blunt instrument 237.20 0.046970297
Cause: Other violence                69.35 0.013732673
Cause: Sports                       126.35 0.025019802
Cause: Falls                        945.75 0.187277228
Cause: Other                        576.50 0.114158416
Mental_Treatment: No               3847.85 0.761950495
Mental_Treatment: Yes              1202.15 0.238049505
Psychiatric_Hospitalisation: No    4593.35 0.909574257
Psychiatric_Hospitalisation: Yes    456.65 0.090425743
Suicide_Lifetime: No               4679.80 0.926693069
Suicide_Lifetime: Yes               370.20 0.073306931
Alcohol_Use: Abstaining            1825.25 0.361435644
Alcohol_Use: Light                 1079.15 0.213693069
Alcohol_Use: Moderate              1404.20 0.278059406
Alcohol_Use: Heavy                  741.40 0.146811881
Drugs: No                          3673.15 0.727356436
Drugs: Yes                         1376.85 0.272643564
```

By TBI aetiology (`Aetiology`)

```
# use the original 'mids' object imp.risk
mi.mean.se.sd.by(imp.risk, "Age", BY = "Aetiology")
```

```
      mean.1      se.1   sd.1   mean.2      se.2     sd.2
Age 33.00919 0.1675076 11.18317 32.64244 0.4472424 10.83849
```

```
# frequencies and proportions
rbind(mi.n.p.by(imp.risk, "Sex", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Race", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Urbanicity", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "BornUS", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Education", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Special_Education", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Employment", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "GCS", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Mental_Treatment", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Psychiatric_Hospitalisation", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Suicide_Lifetime", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Alcohol_Use", BY = "Aetiology"),
      mi.n.p.by(imp.risk, "Drugs", BY = "Aetiology")
      )
```

```
                                       n.1         p.1    n.2         p.2
Sex: Female                        1078.20 0.241757010  82.30 0.139456070
Sex: Male                          3381.65 0.758242990 507.85 0.860543930
Race: White                        2524.80 0.566117695 207.30 0.351266627
Race: Black or AA                   730.05 0.163693846 224.40 0.380242311
Race: Asian / NHPI                  123.10 0.027601825  16.70 0.028297890
Race: Hispanic or Latino            930.55 0.208650515 119.25 0.202067271
Race: AIAN                           16.95 0.003800576   2.45 0.004151487
Race: Non-Hispanic Other             25.85 0.005796159   2.50 0.004236211
Race: More than one race            108.55 0.024339384  17.55 0.029738202
Urbanicity: Rural                  1146.20 0.257004159  87.00 0.147420147
Urbanicity: Urban                  2027.05 0.454510802 350.30 0.593577904
Urbanicity: Suburban               1286.60 0.288485039 152.85 0.259001949
BornUS: Born in the US             3820.00 0.856531049 500.40 0.847920020
BornUS: Born outside the US         639.85 0.143468951  89.75 0.152079980
Education: In school                860.75 0.192999765 169.30 0.286876218
Education: HS/GED                  1698.20 0.380775138 259.30 0.439379819
Education: Some college             906.70 0.203302802  90.70 0.153689740
Education: Associates degree        290.70 0.065181564  30.35 0.051427603
Education: Bachelors degree         520.25 0.116651905  33.30 0.056426332
Education: Graduate degree          183.25 0.041088826   7.20 0.012200288
Special_Education: No              4009.10 0.898931578 512.00 0.867576040
Special_Education: Yes              450.75 0.101068422  78.15 0.132423960
Employment: Competitively employed 3334.80 0.747738153 369.15 0.625518936
Employment: Student                 326.20 0.073141473  41.55 0.070405829
Employment: Retired                 171.10 0.038364519  26.25 0.044480217
Employment: Other employment        108.70 0.024373017  18.70 0.031686859
Employment: Unemployed              519.05 0.116382838 134.50 0.227908159
GCS: Severe                        1658.50 0.371873494 198.20 0.335846819
GCS: Moderate                       482.05 0.108086595  89.60 0.151825087
GCS: Mild                           938.40 0.210410664 181.10 0.306871134
GCS: Sedated                       1370.75 0.307353386 117.25 0.198678302
GCS: Intubated                       10.15 0.002275861   4.00 0.006777938
Mental_Treatment: No               3413.10 0.765294797 434.75 0.736677116
Mental_Treatment: Yes              1046.75 0.234705203 155.40 0.263322884
Psychiatric_Hospitalisation: No    4079.30 0.914672018 514.05 0.871049733
Psychiatric_Hospitalisation: Yes    380.55 0.085327982  76.10 0.128950267
Suicide_Lifetime: No               4157.50 0.932206240 522.30 0.885029230
Suicide_Lifetime: Yes               302.35 0.067793760  67.85 0.114970770
Alcohol_Use: Abstaining            1603.75 0.359597296 221.50 0.375328306
Alcohol_Use: Light                  959.75 0.215197821 119.40 0.202321444
Alcohol_Use: Moderate              1253.30 0.281018420 150.90 0.255697704
Alcohol_Use: Heavy                  643.05 0.144186464  98.35 0.166652546
Drugs: No                          3306.60 0.741415070 366.55 0.621113276
Drugs: Yes                         1153.25 0.258584930 223.60 0.378886724
```

Logistic regression

Linearity assumptions

The sociodemographic factor `Age` is a continuous variable, and therefore the linearity assumption must be explored. Lets assess this before building the logistic regression models using restricted cubic splines.

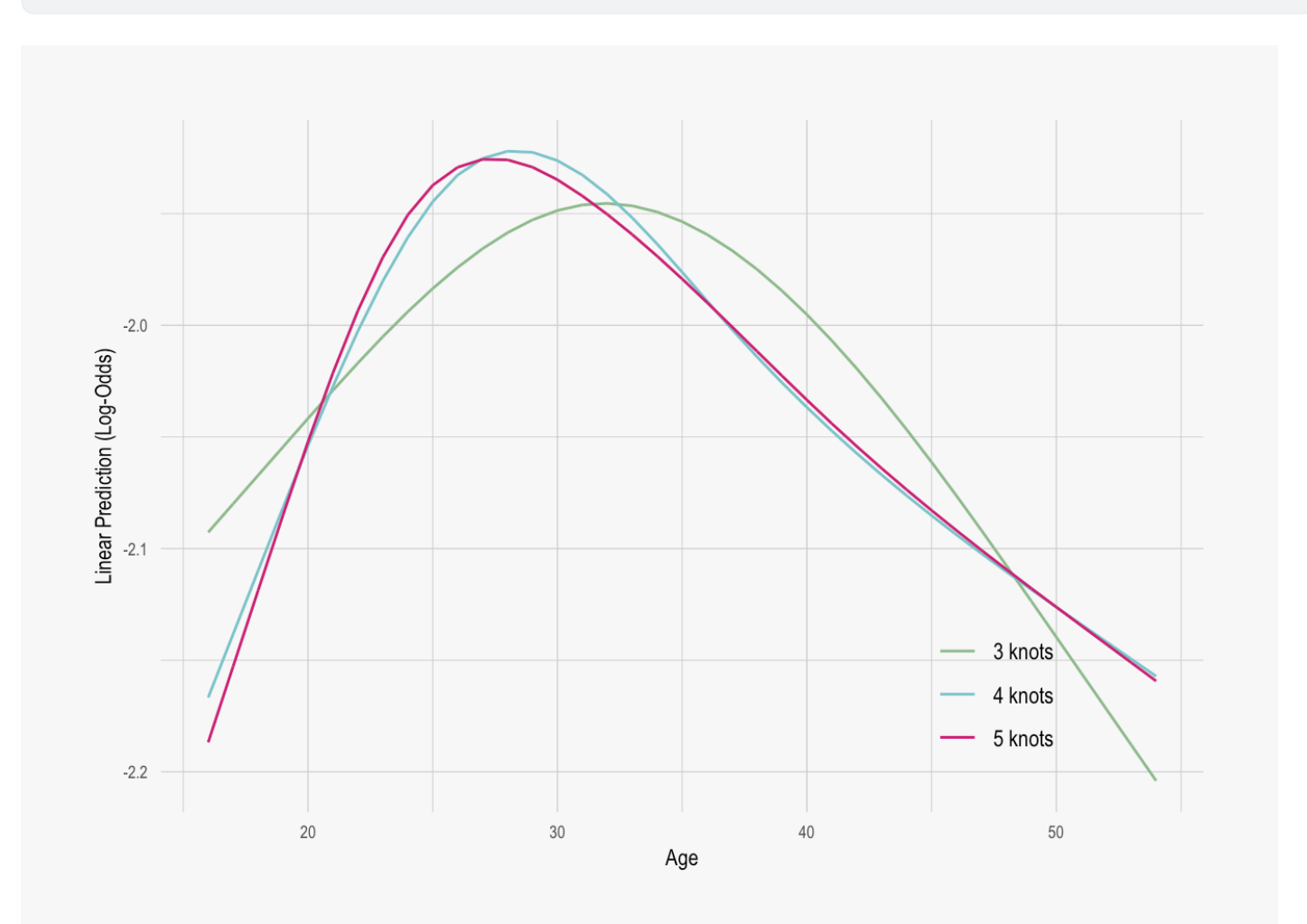
```
# before fitting models, remove missing values
data_risk <- data_risk %>%
  mutate(Aetiology = case_when(
    Cause == "Vehicular" | Cause == "Sports" | Cause == "Falls" | Cause == "Other" ~ 0,
    Cause == "Gunshot wounds" | Cause == "Assault with blunt instrument" | Cause == "Oth
    TRUE ~ NA
  ))

table(data_risk$Aetiology)
```

```

   0    1
4448  588
```

```
baseline_complete <- data_risk[complete.cases(data_risk$Age,
                                              data_risk$Aetiology), ]
```

```
# Now fit models on complete data
agebaseline3_spline <- rcs(baseline_complete$Age, 3)
agebaseline4_spline <- rcs(baseline_complete$Age, 4)
agebaseline5_spline <- rcs(baseline_complete$Age, 5)

# fit logistic regression models
logit_baseline3 <- glm(Aetiology ~ agebaseline3_spline,
                       data = baseline_complete,
                       family = binomial(link = "logit"))

logit_baseline4 <- glm(Aetiology ~ agebaseline4_spline,
                       data = baseline_complete,
                       family = binomial(link = "logit"))

logit_baseline5 <- glm(Aetiology ~ agebaseline5_spline,
                       data = baseline_complete,
                       family = binomial(link = "logit"))

# get linear predictors
lp_baseline3 <- predict(logit_baseline3, type = "link")
lp_baseline4 <- predict(logit_baseline4, type = "link")
lp_baseline5 <- predict(logit_baseline5, type = "link")
```

```
lp_baseline5 <- predict(logit_baseline5, type = "link")

# create data frames for plotting
data_splines <- data.frame(
  age_at_entry = baseline_complete$Age,
  lp_baseline3 = lp_baseline3,
  lp_baseline4 = lp_baseline4,
  lp_baseline5 = lp_baseline5
)

# reshape to long format
data_splines_m <- melt(data_splines, id.vars = 'age_at_entry')

# plot
ggplot(data_splines_m,
                       aes(x=age_at_entry, y=value, colour=variable)) +
  geom_line() +
   scale_colour_manual(labels = c("3 knots", "4 knots", "5 knots"), values = c("darkseag
  theme_brand() +
  labs(x = "Age",
       y = "Linear Prediction (Log-Odds)",
       color = "") +
  theme(legend.position = c(0.8, 0.2))
```



```
model_comparison <- data.frame(
  Model = c("logit_baseline3", "logit_baseline4", "logit_baseline5"),
  Knots = c(3, 4, 5),  # or whatever your knot values are
  AIC = c(
    AIC(logit_baseline3),
    AIC(logit_baseline4),
    AIC(logit_baseline5)
  ),
  BIC = c(
    BIC(logit_baseline3),
    BIC(logit_baseline4),
    BIC(logit_baseline5)
  )
)

model_comparison %>%
  gt() %>%
  tab_header(
    title = "Model Comparison",
    subtitle = "AIC and BIC Values"
  ) %>%
  fmt_number(
    columns = c(AIC, BIC),
    decimals = 2
  ) %>%
  # Highlight minimum AIC in green
  tab_style(
    style = cell_fill(color = "lightblue3"),
    locations = cells_body(
      columns = AIC,
      rows = AIC == min(AIC)
    )
  ) %>%
  # Highlight minimum BIC in green
  tab_style(
    style = cell_fill(color = "lightblue3"),
    locations = cells_body(
      columns = BIC,
      rows = BIC == min(BIC)
    )
  )
```

Model Comparison

AIC and BIC Values

Model	Knots	AIC	BIC
logit_baseline3	3	3,633.66	3,653.23
logit_baseline4	4	3,635.16	3,661.26
logit_baseline5	5	3,637.10	3,669.73

Age at injury should be modelled using restricted cubic splines with 3 knots.

Minimally adjusted models

```
imp.psych_rcs <- with(imp.risk,
                      glm(Aetiology ~ Psychiatric_Hospitalisation + rcs(Age, 3) + Sex, f
```

```
tbl_regression(imp.psych_rcs,
               exponentiate = TRUE) %>% # Shows ORs instead of log-odds
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation"
  )
```

Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation

Variable	**OR**	**95% CI**	**p-value**
Psychiatric_Hospitalisation			
No	—	—	
Yes	1.70	1.30, 2.22	**<0.001**
Age			
rcs(Age, 3)Age	1.01	0.98, 1.03	0.6
rcs(Age, 3)Age'	0.98	0.95, 1.02	0.4
Sex			
Female	—	—	
Male	2.02	1.58, 2.58	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.mntl_rcs <- with(imp.risk,
                     glm(Aetiology ~ Mental_Treatment + rcs(Age, 3) + Sex, family = bin
```

```
tbl_regression(imp.mntl_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Mental Health Treatment"
  )
```

Logistic Regression: TBI Aetiology by Mental Health Treatment

Variable	**OR**	**95% CI**	**p-value**
Mental_Treatment			
No	—	—	
Yes	1.28	1.05, 1.57	**0.016**
Age			
rcs(Age, 3)Age	1.01	0.98, 1.03	0.6
rcs(Age, 3)Age'	0.98	0.95, 1.02	0.4
Sex			
Female	—	—	
Male	2.04	1.59, 2.60	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.suicide_rcs <- with(imp.risk,
                        glm(Aetiology ~ Suicide_Lifetime + rcs(Age, 3) + Sex, family = bin
```

```
tbl_regression(imp.suicide_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Suicide Attempt (Lifetime)"
  )
```

Logistic Regression: TBI Aetiology by Suicide Attempt (Lifetime)

Variable	**OR**	**95% CI**	**p-value**
Suicide_Lifetime			
No	—	—	
Yes	1.92	1.45, 2.55	**<0.001**
Age			
rcs(Age, 3)Age	1.01	0.98, 1.03	0.6
rcs(Age, 3)Age'	0.98	0.95, 1.02	0.4
Sex			
Female	—	—	
Male	2.03	1.59, 2.60	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.drink_rcs <- with(imp.risk,
                      glm(Aetiology ~ Alcohol_Use + rcs(Age, 3) + Sex, family = binomial
```

```
tbl_regression(imp.drink_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Alcohol Use"
  )
```

Logistic Regression: TBI Aetiology by Alcohol Use

Variable	**OR**	**95% CI**	**p-value**
Alcohol_Use			
Abstaining	—	—	
Light	0.91	0.71, 1.17	0.4
Moderate	0.80	0.64, 1.01	0.062
Heavy	1.10	0.84, 1.44	0.5
Age			
rcs(Age, 3)Age	1.01	0.99, 1.03	0.4
rcs(Age, 3)Age'	0.98	0.94, 1.01	0.2
Sex			
Female	—	—	
Male	2.01	1.57, 2.57	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.drug_rcs <- with(imp.risk,
                     glm(Aetiology ~ Drugs + rcs(Age, 3) + Sex, family = binomial))
```

```
tbl_regression(imp.drug_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Illicit Drug Use"
  )
```

Logistic Regression: TBI Aetiology by Illicit Drug Use

Variable	**OR**	**95% CI**	**p-value**
Drugs			
No	—	—	
Yes	1.69	1.41, 2.04	**<0.001**
Age			
rcs(Age, 3)Age	1.01	0.99, 1.03	0.4
rcs(Age, 3)Age'	0.99	0.95, 1.02	0.4
Sex			
Female	—	—	
Male	1.89	1.48, 2.41	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Fully adjusted models

Adjust for all sociodemographic risk factors (age, sex, race, urbanicity, educational attainment, learning difficulties/special education, and employment status) in the logistic regression models.

```
imp.psych.full <- with(imp.risk,
                       glm(Aetiology ~ Psychiatric_Hospitalisation + rcs(Age, 3) + Sex
                           + Race + Urbanicity + Education + Special_Education + Employme
```

```
tbl_regression(imp.psych.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation (Fully Ad
```

Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Psychiatric_Hospitalisation			
No	—	—	
Yes	1.44	1.08, 1.93	**0.013**
Age			
rcs(Age, 3)Age	1.02	1.00, 1.05	0.10
rcs(Age, 3)Age'	0.97	0.93, 1.01	0.11
Sex			
Female	—	—	
Male	1.87	1.45, 2.41	**<0.001**
Race			
White	—	—	
Black or AA	2.85	2.26, 3.59	**<0.001**
Asian / NHPI	1.71	0.97, 3.04	0.064
Hispanic or Latino	1.15	0.88, 1.50	0.3
AIAN	1.26	0.27, 5.84	0.8
Non-Hispanic Other	1.04	0.24, 4.43	>0.9
More than one race	1.88	1.08, 3.28	**0.027**
Urbanicity			
Rural	—	—	
Urban	1.84	1.41, 2.41	**<0.001**
Suburban	1.52	1.14, 2.02	**0.005**
Education			
In school	—	—	
HS/GED	0.83	0.66, 1.04	0.10
Some college	0.56	0.42, 0.75	**<0.001**
Associates degree	0.68	0.44, 1.05	0.078
Bachelors degree	0.42	0.28, 0.63	**<0.001**
Graduate degree	0.26	0.12, 0.58	**<0.001**
Special_Education			
No	—	—	
Yes	0.94	0.71, 1.25	0.7
Employment			
Competitively employed	—	—	
Student	1.20	0.81, 1.79	0.4
Retired	1.09	0.68, 1.74	0.7
Other employment	1.76	1.02, 3.02	**0.041**
Unemployed	1.90	1.50, 2.42	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.mntl.full <- with(imp.risk,
                      glm(Aetiology ~ Mental_Treatment + rcs(Age, 3) + Sex
                          + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(imp.mntl.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Mental Health Treatment (Fully Adjust
```

Logistic Regression: TBI Aetiology by Mental Health Treatment (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Mental_Treatment			
No	—	—	
Yes	1.33	1.07, 1.66	**0.011**
Age			
rcs(Age, 3)Age	1.02	1.00, 1.05	0.10
rcs(Age, 3)Age'	0.97	0.93, 1.01	0.11
Sex			
Female	—	—	
Male	1.90	1.47, 2.46	**<0.001**
Race			
White	—	—	
Black or AA	2.93	2.32, 3.70	**<0.001**
Asian / NHPI	1.76	0.99, 3.11	0.053
Hispanic or Latino	1.18	0.91, 1.55	0.2
AIAN	1.39	0.30, 6.32	0.7
Non-Hispanic Other	1.08	0.25, 4.61	>0.9
More than one race	1.87	1.07, 3.26	**0.028**
Urbanicity			
Rural	—	—	
Urban	1.86	1.42, 2.44	**<0.001**
Suburban	1.53	1.15, 2.04	**0.004**
Education			
In school	—	—	
HS/GED	0.82	0.65, 1.03	0.091
Some college	0.56	0.42, 0.75	**<0.001**
Associates degree	0.67	0.44, 1.04	0.074
Bachelors degree	0.41	0.27, 0.62	**<0.001**
Graduate degree	0.25	0.11, 0.56	**<0.001**
Special_Education			
No	—	—	
Yes	0.94	0.71, 1.25	0.7
Employment			
Competitively employed	—	—	
Student	1.21	0.81, 1.80	0.4
Retired	1.07	0.67, 1.71	0.8
Other employment	1.68	0.97, 2.90	0.062
Unemployed	1.89	1.49, 2.41	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.suicide.full <- with(imp.risk,
                         glm(Aetiology ~ Suicide_Lifetime + rcs(Age, 3) + Sex
                             + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(imp.suicide.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Suicide Attempt (Fully Adjusted Model
```

Logistic Regression: TBI Aetiology by Suicide Attempt (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Suicide_Lifetime			
No	—	—	
Yes	1.85	1.37, 2.51	**<0.001**
Age			
rcs(Age, 3)Age	1.02	0.99, 1.05	0.12
rcs(Age, 3)Age'	0.97	0.93, 1.01	0.13
Sex			
Female	—	—	
Male	1.90	1.47, 2.45	**<0.001**
Race			
White	—	—	
Black or AA	2.91	2.31, 3.67	**<0.001**
Asian / NHPI	1.71	0.97, 3.04	0.064
Hispanic or Latino	1.15	0.88, 1.51	0.3
AIAN	1.21	0.26, 5.63	0.8
Non-Hispanic Other	1.02	0.23, 4.40	>0.9
More than one race	1.82	1.04, 3.18	**0.036**
Urbanicity			
Rural	—	—	
Urban	1.88	1.43, 2.46	**<0.001**
Suburban	1.55	1.16, 2.06	**0.003**
Education			
In school	—	—	
HS/GED	0.84	0.67, 1.05	0.13
Some college	0.57	0.42, 0.76	**<0.001**
Associates degree	0.69	0.45, 1.06	0.092
Bachelors degree	0.43	0.28, 0.65	**<0.001**
Graduate degree	0.26	0.12, 0.58	**0.001**
Special_Education			
No	—	—	
Yes	0.93	0.70, 1.23	0.6
Employment			
Competitively employed	—	—	
Student	1.20	0.81, 1.79	0.4
Retired	1.09	0.68, 1.74	0.7
Other employment	1.76	1.03, 3.03	**0.040**
Unemployed	1.89	1.49, 2.40	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.drink.full <- with(imp.risk,
                       glm(Aetiology ~ Alcohol_Use + rcs(Age, 3) + Sex
                           + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(imp.drink.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Alcohol Use (Fully Adjusted Model)"
  )
```

Logistic Regression: TBI Aetiology by Alcohol Use (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Alcohol_Use			
Abstaining	—	—	
Light	1.09	0.84, 1.41	0.5
Moderate	1.03	0.81, 1.31	0.8
Heavy	1.29	0.97, 1.72	0.077
Age			
rcs(Age, 3)Age	1.02	1.0, 1.05	0.12
rcs(Age, 3)Age'	0.97	0.93, 1.01	0.12
Sex			
Female	—	—	
Male	1.84	1.43, 2.38	**<0.001**
Race			
White	—	—	
Black or AA	2.86	2.26, 3.61	**<0.001**
Asian / NHPI	1.72	0.98, 3.05	0.061
Hispanic or Latino	1.16	0.89, 1.52	0.3
AIAN	1.42	0.31, 6.48	0.6
Non-Hispanic Other	1.04	0.24, 4.43	>0.9
More than one race	1.87	1.07, 3.26	**0.028**
Urbanicity			
Rural	—	—	
Urban	1.89	1.44, 2.47	**<0.001**
Suburban	1.55	1.16, 2.07	**0.003**
Education			
In school	—	—	
HS/GED	0.84	0.67, 1.06	0.14
Some college	0.57	0.43, 0.77	**<0.001**
Associates degree	0.71	0.46, 1.10	0.13
Bachelors degree	0.42	0.28, 0.65	**<0.001**
Graduate degree	0.27	0.12, 0.59	**0.001**
Special_Education			
No	—	—	
Yes	0.98	0.74, 1.30	0.9
Employment			
Competitively employed	—	—	
Student	1.24	0.83, 1.85	0.3
Retired	1.18	0.74, 1.87	0.5
Other employment	1.82	1.06, 3.13	**0.031**
Unemployed	2.00	1.58, 2.54	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
imp.drug.full <- with(imp.risk,
                      glm(Aetiology ~ Drugs + rcs(Age, 3) + Sex
                          + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(imp.drug.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Illicit Drug Use (Fully Adjusted Mode
```

Logistic Regression: TBI Aetiology by Illicit Drug Use (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Drugs			
No	—	—	
Yes	1.56	1.28, 1.90	**<0.001**
Age			
rcs(Age, 3)Age	1.03	1.00, 1.05	0.064
rcs(Age, 3)Age'	0.97	0.93, 1.01	0.12
Sex			
Female	—	—	
Male	1.78	1.38, 2.30	**<0.001**

Race			
White	—	—	
Black or AA	2.92	2.31, 3.68	**<0.001**
Asian / NHPI	1.84	1.04, 3.26	**0.038**
Hispanic or Latino	1.21	0.93, 1.59	0.2
AIAN	1.36	0.30, 6.20	0.7
Non-Hispanic Other	1.10	0.26, 4.70	0.9
More than one race	1.84	1.05, 3.23	**0.033**
Urbanicity			
Rural	—	—	
Urban	1.84	1.40, 2.41	**<0.001**
Suburban	1.54	1.15, 2.05	**0.003**
Education			
In school	—	—	
HS/GED	0.83	0.66, 1.05	0.12
Some college	0.57	0.43, 0.76	**<0.001**
Associates degree	0.70	0.45, 1.08	0.10
Bachelors degree	0.44	0.29, 0.67	**<0.001**
Graduate degree	0.27	0.12, 0.60	**0.001**
Special_Education			
No	—	—	
Yes	0.94	0.71, 1.25	0.7
Employment			
Competitively employed	—	—	
Student	1.26	0.85, 1.88	0.2
Retired	1.11	0.70, 1.77	0.7
Other employment	1.80	1.05, 3.10	**0.033**
Unemployed	1.87	1.47, 2.37	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Assessing for multicollinearity

An important step in the logistic regression analysis is to assess the assumption of no strong multicollinearity. To do so, we used the `car::vif()` function to calculate the variance inflation factor (VIF), which was pooled across all imputed datasets.

For the `car::vif()` output, the generalized VIF is found in the GVIF column, and the GVIF^(1/(2*Df)) column is the adjusted generalized standard error inflation factor (aGSIF). It is recommended that the aGSIF value to evaluate collinearity, since it allows predictors with different Df (different number of terms in the model) to be comparable (Nahhas, 2024).

For aGSIFs, our rule of thumb for what is large is that values **above 1.6 may be of concern** and values above 2.2 or 3.2 are indicative of a more serious problem (Nahhas, 2024).

Minimally adjusted models

Psychiatric Hospitalisation

```
vif_results_psy <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.psych_rcs <- glm(Aetiology ~ Psychiatric_Hospitalisation + rcs(Age, 3) + Sex,
                  data = complete_data,
                  family = binomial)
  car::vif(imp.psych_rcs)
})

# average VIF across imputations
vif_averaged1 <- Reduce("+", vif_results_psy) / length(vif_results_psy)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged1)
```

```
                                GVIF Df GVIF^(1/(2*Df))
Psychiatric_Hospitalisation 1.008975  1        1.004477
rcs(Age, 3)                 1.005628  2        1.001404
Sex                         1.009415  1        1.004696
```

Mental Health Treatment

```
vif_results_mnt <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.mntl_rcs <- glm(Aetiology ~ Mental_Treatment + rcs(Age, 3) + Sex,
                  data = complete_data,
                  family = binomial)
  car::vif(imp.mntl_rcs)
})

# average VIF across imputations
vif_averaged2 <- Reduce("+", vif_results_mnt) / length(vif_results_mnt)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged2)
```

```
                     GVIF Df GVIF^(1/(2*Df))
Mental_Treatment 1.029334  1        1.014561
rcs(Age, 3)      1.013826  2        1.003439
Sex              1.022122  1        1.011001
```

Suicide Attempt

```
vif_results_suicide <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.suicide_rcs <- glm(Aetiology ~ Suicide_Lifetime + rcs(Age, 3) + Sex,
                  data = complete_data,
                  family = binomial)
  car::vif(imp.suicide_rcs)
})

# average VIF across imputations
vif_averaged3 <- Reduce("+", vif_results_suicide) / length(vif_results_suicide)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged3)
```

```
                     GVIF Df GVIF^(1/(2*Df))
Suicide_Lifetime 1.009917  1        1.004946
rcs(Age, 3)      1.005770  2        1.001439
Sex              1.010997  1        1.005484
```

Alcohol Use

```
vif_results_drink <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.drink_rcs <- glm(Aetiology ~ Alcohol_Use + rcs(Age, 3) + Sex,
                  data = complete_data,
                  family = binomial)
  car::vif(imp.drink_rcs)
})

# average VIF across imputations
vif_averaged4 <- Reduce("+", vif_results_drink) / length(vif_results_drink)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged4)
```

```
                GVIF Df GVIF^(1/(2*Df))
Alcohol_Use 1.043281  3        1.007087
rcs(Age, 3) 1.031944  2        1.007892
Sex         1.015954  1        1.007946
```

Illicit Drug Use

```
vif_results_drugs <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.drug_rcs <- glm(Aetiology ~ Drugs + rcs(Age, 3) + Sex,
                  data = complete_data,
                  family = binomial)
  car::vif(imp.drug_rcs)
})

# average VIF across imputations
vif_averaged5 <- Reduce("+", vif_results_drugs) / length(vif_results_drugs)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged5)
```

```
                GVIF Df GVIF^(1/(2*Df))
Drugs       1.033230  1        1.016479
rcs(Age, 3) 1.034091  2        1.008416
Sex         1.005504  1        1.002748
```

Fully adjusted models

Psychiatric Hospitalisation

```
vif_results_psy_full <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.psych.full <- glm(Aetiology ~ Psychiatric_Hospitalisation + rcs(Age, 3) + Sex
                       + Race + Urbanicity + Education + Special_Education + Employme
                  data = complete_data,
                  family = binomial)
  car::vif(imp.psych.full)
})

# average VIF across imputations
vif_averaged6 <- Reduce("+", vif_results_psy_full) / length(vif_results_psy_full)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged6)
```

```
                                GVIF Df GVIF^(1/(2*Df))
Psychiatric_Hospitalisation 1.098130  1        1.047916
rcs(Age, 3)                 1.371583  2        1.082194
Sex                         1.047931  1        1.023685
Race                        1.296136  6        1.021850
Urbanicity                  1.154266  2        1.036516
Education                   1.263006  5        1.023624
Special_Education           1.084070  1        1.041186
Employment                  1.566887  4        1.057740
```

Mental Health Treatment

```
vif_results_mntl_full <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.mntl.full <- glm(Aetiology ~ Mental_Treatment + rcs(Age, 3) + Sex
                       + Race + Urbanicity + Education + Special_Education + Employme
                  data = complete_data,
                  family = binomial)
  car::vif(imp.mntl.full)
})

# average VIF across imputations
vif_averaged7 <- Reduce("+", vif_results_mntl_full) / length(vif_results_mntl_full)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged7)
```

```
                      GVIF Df GVIF^(1/(2*Df))
Mental_Treatment  1.161683  1        1.077814
rcs(Age, 3)       1.378351  2        1.083526
Sex               1.057923  1        1.028553
Race              1.323506  6        1.023631
Urbanicity        1.152580  2        1.036137
Education         1.267786  5        1.024010
Special_Education 1.084990  1        1.041628
Employment        1.586520  4        1.059388
```

Suicide Attempt

```
vif_results_suicide_full <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.suicide.full <- glm(Aetiology ~ Suicide_Lifetime + rcs(Age, 3) + Sex
                       + Race + Urbanicity + Education + Special_Education + Employme
                  data = complete_data,
                  family = binomial)
  car::vif(imp.suicide.full)
})

# average VIF across imputations
vif_averaged8 <- Reduce("+", vif_results_suicide_full) / length(vif_results_suicide_full
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged8)
```

```
                      GVIF Df GVIF^(1/(2*Df))
Suicide_Lifetime  1.080531  1        1.039485
rcs(Age, 3)       1.375115  2        1.082890
Sex               1.049081  1        1.024210
Race              1.307672  6        1.022605
Urbanicity        1.152295  2        1.036073
Education         1.265689  5        1.023841
Special_Education 1.080675  1        1.039555
Employment        1.538134  4        1.055294
```

Alcohol Use

```
vif_results_drink_full <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.drink.full <- glm(Aetiology ~ Alcohol_Use + rcs(Age, 3) + Sex
                       + Race + Urbanicity + Education + Special_Education + Employme
                  data = complete_data,
                  family = binomial)
  car::vif(imp.drink.full)
})

# average VIF across imputations
vif_averaged9 <- Reduce("+", vif_results_drink_full) / length(vif_results_drink_full)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged9)
```

```
                      GVIF Df GVIF^(1/(2*Df))
Alcohol_Use       1.126479  3        1.020047
rcs(Age, 3)       1.392005  2        1.086215
Sex               1.057037  1        1.028123
Race              1.318836  6        1.023329
Urbanicity        1.154451  2        1.036557
Education         1.284881  5        1.025383
Special_Education 1.067904  1        1.033394
Employment        1.527043  4        1.054340
```

Illicit Drug Use

```
vif_results_drugs_full <- lapply(1:imp.risk$m, function(i) {
  complete_data <- complete(imp.risk, i)
  imp.drugs.full <- glm(Aetiology ~ Drugs + rcs(Age, 3) + Sex
                       + Race + Urbanicity + Education + Special_Education + Employme
                  data = complete_data,
                  family = binomial)
  car::vif(imp.drugs.full)
})

# average VIF across imputations
vif_averaged10 <- Reduce("+", vif_results_drugs_full) / length(vif_results_drugs_full)
print("Average VIF across imputations:")
```

```
[1] "Average VIF across imputations:"
```

```
print(vif_averaged10)
```

```
                      GVIF Df GVIF^(1/(2*Df))
Drugs             1.095217  1        1.046526
rcs(Age, 3)       1.412512  2        1.090178
Sex               1.043489  1        1.021513
Race              1.305435  6        1.022459
Urbanicity        1.151604  2        1.035917
Education         1.265127  5        1.023767
Special_Education 1.072526  1        1.035628
Employment        1.530157  4        1.054609
```

Sensitivity analysis

Are similar results obtained when the analysis is restricted to the complete cases only?

```
data_complete <- data_risk
```

```
data_complete <- data_complete[complete.cases(data_complete[ , c("Aetiology", "Age", "Se
```

```
data_complete %>%
  distinct(Mod1Id) %>%
  nrow()
```

```
[1] 4186
```

Descriptive statistics

```
data_complete %>%
  tbl_summary(by = Aetiology, include = c(Age, Sex, Race, BornUS, Urbanicity, Employment
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
  test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  add_overall()
```

Characteristic	Overall N = 4,186[1]	0 N = 3,719[1]	1 N = 467[1]	p-value[2]
Age	31 (23, 42)	31 (23, 42)	30 (23, 41)	0.8
Sex				<0.001
Female	987 (24%)	925 (25%)	62 (13%)	
Male	3,199 (76%)	2,794 (75%)	405 (87%)	
Race				<0.001
White	2,288 (55%)	2,119 (57%)	169 (36%)	
Black or AA	780 (19%)	602 (16%)	178 (38%)	
Asian / NHPI	110 (2.6%)	96 (2.6%)	14 (3.0%)	
Hispanic or Latino	866 (21%)	777 (21%)	89 (19%)	
AIAN	16 (0.4%)	14 (0.4%)	2 (0.4%)	
Non-Hispanic Other	21 (0.5%)	20 (0.5%)	1 (0.2%)	
More than one race	105 (2.5%)	91 (2.4%)	14 (3.0%)	
BornUS				0.9
Born in the US	3,633 (87%)	3,229 (87%)	404 (87%)	
Born outside the US	553 (13%)	490 (13%)	63 (13%)	
Urbanicity				<0.001
Rural	1,042 (25%)	972 (26%)	70 (15%)	
Urban	1,924 (46%)	1,651 (44%)	273 (58%)	
Suburban	1,220 (29%)	1,096 (29%)	124 (27%)	
Employment				<0.001
Competitively employed	3,102 (74%)	2,806 (75%)	296 (63%)	
Student	318 (7.6%)	287 (7.7%)	31 (6.6%)	
Retired	162 (3.9%)	140 (3.8%)	22 (4.7%)	
Other employment	101 (2.4%)	88 (2.4%)	13 (2.8%)	
Unemployed	503 (12%)	398 (11%)	105 (22%)	
Education				<0.001
In school	831 (20%)	706 (19%)	125 (27%)	
HS/GED	1,599 (38%)	1,392 (37%)	207 (44%)	
Some college	852 (20%)	778 (21%)	74 (16%)	
Associates degree	279 (6.7%)	252 (6.8%)	27 (5.8%)	
Bachelors degree	461 (11%)	432 (12%)	29 (6.2%)	
Graduate degree	164 (3.9%)	159 (4.3%)	5 (1.1%)	
Special_Education	426 (10%)	362 (9.7%)	64 (14%)	0.011
Psychiatric_Hospitalisation	346 (8.3%)	288 (7.7%)	58 (12%)	<0.001
Mental_Treatment	980 (23%)	856 (23%)	124 (27%)	0.095
Suicide_Lifetime	279 (6.7%)	227 (6.1%)	52 (11%)	<0.001
Alcohol_Use				0.3
Abstaining	1,523 (36%)	1,342 (36%)	181 (39%)	
Light	905 (22%)	813 (22%)	92 (20%)	
Moderate	1,161 (28%)	1,042 (28%)	119 (25%)	
Heavy	597 (14%)	522 (14%)	75 (16%)	
Drugs	1,109 (26%)	930 (25%)	179 (38%)	<0.001

[1] Median (Q1, Q3); n (%)

[2] Wilcoxon rank sum test; Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Aetiology by psychiatric hospitalisation

```
data_complete %>%
  tbl_summary(by = Psychiatric_Hospitalisation, include = Aetiology) %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
  test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  add_overall()
```

Characteristic	Overall N = 4,186[1]	No N = 3,840[1]	Yes N = 346[1]	p-value[2]
Aetiology	467 (11%)	409 (11%)	58 (17%)	<0.001

[1] n (%)

[2] Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Aetiology by mental health treatment

```
data_complete %>%
  tbl_summary(by = Mental_Treatment, include = Aetiology) %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
  test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  add_overall()
```

Characteristic	Overall N = 4,186[1]	No N = 3,206[1]	Yes N = 980[1]	p-value[2]
Aetiology	467 (11%)	343 (11%)	124 (13%)	0.10

[1] n (%)

[2] Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Aetiology by suicide attempt

```
data_complete %>%
  tbl_summary(by = Suicide_Lifetime, include = Aetiology) %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
  test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  add_overall()
```

Characteristic	Overall N = 4,186[1]	No N = 3,907[1]	Yes N = 279[1]	p-value[2]
Aetiology	467 (11%)	415 (11%)	52 (19%)	<0.001

[1] n (%)

[2] Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Aetiology by alcohol use

```
data_complete %>%
  tbl_summary(by = Alcohol_Use, include = Aetiology) %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
  test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  add_overall()
```

Characteristic	Overall N = 4,186[1]	Abstaining N = 1,523[1]	Light N = 905[1]	Moderate N = 1,161[1]	Heavy N = 597[1]	p-value[2]
Aetiology	467 (11%)	181 (12%)	92 (10%)	119 (10%)	75 (13%)	0.3

[1] n (%)

[2] Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Aetiology by illicit drug use

```
data_complete %>%
  tbl_summary(by = Drugs, include = Aetiology) %>%
  add_p(
    # perform  chi-square with simulation to avoid computational constraints
    test = all_categorical() ~ "chisq.test",
  test.args = all_categorical() ~ list(simulate.p.value = TRUE)) %>%
  add_overall()
```

Characteristic	Overall N = 4,186[1]	No N = 3,077[1]	Yes N = 1,109[1]	p-value[2]
Aetiology	467 (11%)	288 (9.4%)	179 (16%)	<0.001

[1] n (%)

[2] Pearson's Chi-squared test with simulated p-value (based on 2000 replicates)

Minimally adjusted models

```
cc.psych_rcs <- glm(Aetiology ~ Psychiatric_Hospitalisation + rcs(Age, 3) + Sex, family
```

```
tbl_regression(cc.psych_rcs,
               exponentiate = TRUE) %>% # Shows ORs instead of log-odds
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation"
  )
```

Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation

Variable	OR	95% CI	p-value
Psychiatric_Hospitalisation			
No	—	—	
Yes	1.83	1.34, 2.46	**<0.001**
Age			
rcs(Age, 3)Age	1.00	0.98, 1.03	0.9
rcs(Age, 3)Age'	0.99	0.95, 1.03	0.7
Sex			
Female	—	—	
Male	2.24	1.70, 2.98	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
cc.mntl_rcs <- glm(Aetiology ~ Mental_Treatment + rcs(Age, 3) + Sex, family = binomial,
```

```
tbl_regression(cc.mntl_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Mental Health Treatment"
  )
```

Logistic Regression: TBI Aetiology by Mental Health Treatment

Variable	OR	95% CI	p-value
Mental_Treatment			
No	—	—	
Yes	1.34	1.07, 1.67	**0.010**
Age			
rcs(Age, 3)Age	1.00	0.97, 1.03	>0.9
rcs(Age, 3)Age'	0.99	0.95, 1.03	0.7
Sex			
Female	—	—	
Male	2.26	1.72, 3.01	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
cc.suicide_rcs <- glm(Aetiology ~ Suicide_Lifetime + rcs(Age, 3) + Sex, family = binomia
```

```
tbl_regression(cc.suicide_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Suicide Attempt"
  )
```

Logistic Regression: TBI Aetiology by Suicide Attempt

Variable	OR	95% CI	p-value
Suicide_Lifetime			
No	—	—	
Yes	2.10	1.51, 2.88	**<0.001**
Age			
rcs(Age, 3)Age	1.00	0.98, 1.03	0.9
rcs(Age, 3)Age'	0.99	0.95, 1.03	0.7
Sex			
Female	—	—	
Male	2.25	1.71, 3.00	**<0.001**

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

```
cc.drink_rcs <- glm(Aetiology ~ Alcohol_Use + rcs(Age, 3) + Sex, family = binomial, data
```

```
tbl_regression(cc.drink_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Alcohol Use"
  )
```

Logistic Regression: TBI Aetiology by Alcohol Use

Variable	OR	95% CI	p-value

Variable	**OR**	**95% CI**	**p-value**
Alcohol_Use			
Abstaining	—	—	
Light	0.85	0.65, 1.11	0.2
Moderate	0.78	0.60, 1.00	**0.047**
Heavy	1.06	0.79, 1.41	0.7
Age			
rcs(Age, 3)Age	1.01	0.98, 1.03	0.6
rcs(Age, 3)Age'	0.99	0.95, 1.03	0.5
Sex			
Female	—	—	
Male	2.21	1.68, 2.95	**<0.001**
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

```
cc.drug_rcs <- glm(Aetiology ~ Drugs + rcs(Age, 3) + Sex, family = binomial, data = data
```

```
tbl_regression(cc.drug_rcs,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Illicit Drug Use"
  )
```

Logistic Regression: TBI Aetiology by Illicit Drug Use

Variable	**OR**	**95% CI**	**p-value**
Drugs			
No	—	—	
Yes	1.81	1.48, 2.23	**<0.001**
Age			
rcs(Age, 3)Age	1.00	0.98, 1.03	0.7
rcs(Age, 3)Age'	1.00	0.96, 1.04	0.8
Sex			
Female	—	—	
Male	2.08	1.58, 2.77	**<0.001**
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Fully adjusted models

Adjust for all sociodemographic risk factors (age, sex, race, urbanicity, educational attainment, learning difficulties/special education, and employment status) in the logistic regression models.

```
cc.psych.full <- glm(Aetiology ~ Psychiatric_Hospitalisation + rcs(Age, 3) + Sex
                         + Race + Urbanicity + Education + Special_Education + Employme
```

```
tbl_regression(cc.psych.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation (Fully Ad
  )
```

Logistic Regression: TBI Aetiology by Psychiatric Hospitalisation (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Psychiatric_Hospitalisation			
No	—	—	
Yes	1.50	1.07, 2.07	**0.016**
Age			
rcs(Age, 3)Age	1.02	0.99, 1.05	0.3
rcs(Age, 3)Age'	0.98	0.94, 1.02	0.4
Sex			
Female	—	—	
Male	2.03	1.53, 2.74	**<0.001**
Race			
White	—	—	
Black or AA	2.83	2.21, 3.63	**<0.001**
Asian / NHPI	1.96	1.03, 3.48	**0.030**
Hispanic or Latino	1.08	0.80, 1.44	0.6
AIAN	1.24	0.19, 4.79	0.8
Non-Hispanic Other	0.60	0.03, 2.97	0.6
More than one race	1.91	1.01, 3.36	**0.034**
Urbanicity			
Rural	—	—	
Urban	1.89	1.42, 2.56	**<0.001**
Suburban	1.58	1.15, 2.17	**0.005**
Education			
In school	—	—	
HS/GED	0.90	0.69, 1.17	0.4
Some college	0.60	0.43, 0.83	**0.002**
Associates degree	0.77	0.47, 1.21	0.3
Bachelors degree	0.50	0.31, 0.77	**0.003**
Graduate degree	0.23	0.08, 0.53	**0.002**
Special_Education			
No	—	—	
Yes	1.01	0.74, 1.37	>0.9
Employment			
Competitively employed	—	—	
Student	1.06	0.67, 1.64	0.8
Retired	1.11	0.65, 1.83	0.7
Other employment	1.57	0.80, 2.84	0.2
Unemployed	2.11	1.61, 2.74	**<0.001**
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

```
cc.mntl.full <- glm(Aetiology ~ Mental_Treatment + rcs(Age, 3) + Sex
                        + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(cc.mntl.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Mental Health Treatment (Fully Adjust
  )
```

Logistic Regression: TBI Aetiology by Mental Health Treatment (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Mental_Treatment			
No	—	—	
Yes	1.34	1.05, 1.72	**0.018**
Age			
rcs(Age, 3)Age	1.01	0.98, 1.05	0.3
rcs(Age, 3)Age'	0.98	0.94, 1.02	0.4
Sex			
Female	—	—	
Male	2.06	1.55, 2.77	**<0.001**
Race			
White	—	—	
Black or AA	2.91	2.27, 3.74	**<0.001**
Asian / NHPI	2.00	1.05, 3.57	**0.025**
Hispanic or Latino	1.11	0.82, 1.49	0.5
AIAN	1.37	0.21, 5.21	0.7
Non-Hispanic Other	0.62	0.03, 3.09	0.6
More than one race	1.88	0.99, 3.31	**0.039**
Urbanicity			
Rural	—	—	
Urban	1.91	1.43, 2.58	**<0.001**
Suburban	1.59	1.16, 2.19	**0.004**
Education			
In school	—	—	
HS/GED	0.89	0.69, 1.16	0.4
Some college	0.59	0.43, 0.82	**0.002**
Associates degree	0.76	0.47, 1.20	0.3
Bachelors degree	0.49	0.30, 0.76	**0.002**
Graduate degree	0.22	0.08, 0.52	**0.002**
Special_Education			
No	—	—	
Yes	1.01	0.73, 1.37	>0.9
Employment			
Competitively employed	—	—	
Student	1.06	0.67, 1.65	0.8
Retired	1.11	0.65, 1.82	0.7
Other employment	1.47	0.75, 2.69	0.2
Unemployed	2.10	1.60, 2.74	**<0.001**
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

```
cc.suicide.full <- glm(Aetiology ~ Suicide_Lifetime + rcs(Age, 3) + Sex
                        + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(cc.suicide.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Suicide Attempt (Fully Adjusted Model
  )
```

Logistic Regression: TBI Aetiology by Suicide Attempt (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Suicide_Lifetime			
No	—	—	
Yes	1.99	1.39, 2.79	**<0.001**
Age			
rcs(Age, 3)Age	1.01	0.98, 1.05	0.4
rcs(Age, 3)Age'	0.98	0.94, 1.03	0.4
Sex			
Female	—	—	
Male	2.06	1.55, 2.78	**<0.001**
Race			
White	—	—	
Black or AA	2.90	2.26, 3.71	**<0.001**
Asian / NHPI	1.95	1.02, 3.47	**0.031**
Hispanic or Latino	1.08	0.80, 1.45	0.6
AIAN	1.15	0.17, 4.51	0.9
Non-Hispanic Other	0.62	0.03, 3.09	0.6
More than one race	1.81	0.95, 3.20	0.054
Urbanicity			
Rural	—	—	
Urban	1.94	1.45, 2.62	**<0.001**
Suburban	1.61	1.18, 2.21	**0.003**
Education			
In school	—	—	
HS/GED	0.91	0.70, 1.18	0.5
Some college	0.60	0.43, 0.83	**0.002**
Associates degree	0.78	0.48, 1.23	0.3
Bachelors degree	0.51	0.32, 0.80	**0.004**
Graduate degree	0.23	0.08, 0.53	**0.002**
Special_Education			
No	—	—	
Yes	0.98	0.71, 1.33	0.9
Employment			
Competitively employed	—	—	
Student	1.06	0.67, 1.65	0.8
Retired	1.11	0.65, 1.82	0.7
Other employment	1.58	0.81, 2.87	0.2
Unemployed	2.09	1.60, 2.72	**<0.001**
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

```
cc.drink.full <- glm(Aetiology ~ Alcohol_Use + rcs(Age, 3) + Sex
                        + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(cc.drink.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Alcohol Use (Fully Adjusted Model)"
  )
```

Logistic Regression: TBI Aetiology by Alcohol Use (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Alcohol_Use			
Abstaining	—	—	
Light	1.01	0.76, 1.33	>0.9
Moderate	0.97	0.74, 1.25	0.8
Heavy	1.21	0.89, 1.64	0.2
Age			
rcs(Age, 3)Age	1.02	0.98, 1.05	0.3
rcs(Age, 3)Age'	0.98	0.94, 1.02	0.4
Sex			
Female	—	—	
Male	2.00	1.51, 2.70	**<0.001**
Race			
White	—	—	
Black or AA	2.82	2.20, 3.61	**<0.001**
Asian / NHPI	1.94	1.02, 3.45	**0.031**
Hispanic or Latino	1.08	0.80, 1.45	0.6
AIAN	1.40	0.21, 5.34	0.7
Non-Hispanic Other	0.60	0.03, 2.98	0.6
More than one race	1.88	1.00, 3.32	**0.038**
Urbanicity			
Rural	—	—	
Urban	1.94	1.45, 2.62	**<0.001**
Suburban	1.60	1.17, 2.20	**0.004**
Education			
In school	—	—	
HS/GED	0.92	0.71, 1.19	0.5
Some college	0.61	0.44, 0.85	**0.003**
Associates degree	0.81	0.50, 1.28	0.4
Bachelors degree	0.51	0.32, 0.80	**0.004**
Graduate degree	0.24	0.08, 0.55	**0.002**
Special_Education			
No	—	—	
Yes	1.05	0.77, 1.42	0.7
Employment			
Competitively employed	—	—	
Student	1.08	0.68, 1.67	0.7
Retired	1.22	0.72, 1.98	0.4
Other employment	1.60	0.82, 2.92	0.14
Unemployed	2.21	1.69, 2.87	**<0.001**
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

```
cc.drug.full <- glm(Aetiology ~ Drugs + rcs(Age, 3) + Sex
                        + Race + Urbanicity + Education + Special_Education + Employm
```

```
tbl_regression(cc.drug.full,
               exponentiate = TRUE) %>%
  bold_p() %>%
  bold_labels() %>%
  modify_header(label = "**Variable**") %>%
  as_gt() %>%  # Convert to gt for further customization
  gt::tab_header(
    title = "Logistic Regression: TBI Aetiology by Illicit Drug Use (Fully Adjusted Mode
  )
```

Logistic Regression: TBI Aetiology by Illicit Drug Use (Fully Adjusted Model)

Variable	**OR**	**95% CI**	**p-value**
Drugs			
No	—	—	
Yes	1.64	1.32, 2.04	**<0.001**
Age			
rcs(Age, 3)Age	1.02	0.99, 1.05	0.2
rcs(Age, 3)Age'	0.98	0.94, 1.03	0.4
Sex			
Female	—	—	
Male	1.93	1.45, 2.59	**<0.001**
Race			
White	—	—	
Black or AA	2.92	2.28, 3.74	**<0.001**
Asian / NHPI	2.09	1.10, 3.73	**0.017**
Hispanic or Latino	1.15	0.85, 1.54	0.4
AIAN	1.36	0.21, 5.22	0.7
Non-Hispanic Other	0.63	0.03, 3.15	0.7
More than one race	1.87	0.99, 3.31	**0.041**
Urbanicity			
Rural	—	—	
Urban	1.87	1.40, 2.53	**<0.001**
Suburban	1.59	1.17, 2.19	**0.004**
Education			
In school	—	—	
HS/GED	0.91	0.70, 1.18	0.5
Some college	0.61	0.44, 0.84	**0.003**
Associates degree	0.79	0.49, 1.25	0.3
Bachelors degree	0.52	0.33, 0.82	**0.005**
Graduate degree	0.24	0.08, 0.56	**0.003**
Special_Education			
No	—	—	
Yes	1.01	0.74, 1.37	>0.9
Employment			
Competitively employed	—	—	
Student	1.12	0.71, 1.74	0.6
Retired	1.15	0.67, 1.87	0.6
Other employment	1.60	0.82, 2.92	0.14
Unemployed	2.05	1.56, 2.66	**<0.001**
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Clean up

Save data to the .RProject.

```
save(object = longitudinal_complete, file = "data/longitudinal_complete.RData") # save t
save(object = longitudinal_clean, file = "data/longitudinal_clean.RData") # cleaned data
save(object = data_risk, file = "data/data_risk.RData") # data use for the psychiatric r
```

Clean environment.

```
gc() # garabage, free unused R memory
rm(list = ls()) # remove all from R environment
cat("\014") # clear console
```